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Right answers, wrong moves: moral psychology as a test of language models

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Contents

Abstract
Main
Results
- Three schools, one method
- Recognition saturates; pressure separates
- The gate carries the floor
- Both poles are instrumented; one is where models fail
- Saying and doing come apart, in a school-specific way
- Per-model profiles and what capability explains
- What repeats, and what the judges add

Discussion
Methods
Data and code availability
Author contributions, competing interests, acknowledgements
References
Figure legends
Extended Data
- Extended Data Figures
- Extended Data Tables

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Abstract

Benchmarks of the values of language models mostly test stated answers, pool unrelated moral theories into one score, or rest on a judge whose grades cannot be taken apart. We built a benchmark that does none of these, from three schools of moral psychology, Moral Foundations Theory, Schwartz's refined values and the neo-Kohlbergian schema tradition, each kept under its own construct: not which values a model holds, which the theories say is not rankable, but whether it can operate a theory's descriptive apparatus faithfully when asked, graded by the theory's own rules against its caricature. Items were authored under a mechanical law, administered on a ladder from keyed recognition through sustained argument to an agentic episode whose action and written reasons are scored blind to each other, and judged out of vendor family with an evidence quote for every verdict. Across fifteen models, 9,776 administrations and 24,518 verdicts from six judges, recognition saturated at 1.000 for every model; sustained pressure separated models in every panel under every judge (η^2 0.16–0.42 on the board, 0.15–0.55 on a nine-model frame, 0.22–0.67 on 160 fresh items; permutation $p \le 0.0025$ throughout), with the discrimination carried by anti-caricature gates that tripped concordantly across judges; and in 13–28 % of agentic episodes a model wrote a faithful analysis and then took a compromising action, against a reverse discordance of 6 % or less, a gap that differed by school and fell to 4–5 % when the competing pressure was removed. The three schools' model means correlate 0.41–0.81 across models, and the one column that published capability does not order, Schwartz agentic behaviour (ρ 0.03, 95 % CI −0.65 to 0.68), is the one a pooled score would have averaged away.

Main

Three facts shape any attempt to measure the values of a language model. Stated tests have saturated: on fresh multiple-choice items whose options are analyses rather than verdicts, with named caricatures of the theory as distractors, every model we measured answered every item correctly, and the saturation of keyed moral benchmarks is now reported across the field. Benchmarks that do separate models mostly do so with a language-model judge whose score is rarely decomposed, so a difference of 0.2 between two models cannot be traced to a construct, a judge's habit or the length of a response. And benchmarks of "values" routinely commit two conflations that moral psychology itself refuses: they flatten distinct theories into one number, and they blend the question *can this model reason faithfully inside a named theory when asked* with the question *which values does its unprompted behaviour express*. Moral psychology is descriptive. None of its theories says that a model that weights Loyalty over Care, or that argues from maintaining norms rather than from principle, is worse. What the theories do license is a competence: whether the theory's own descriptive apparatus can be operated faithfully on request, as the theory's scholars operate it when they code a vignette, and whether that competence survives pressure.

The distinction matters to psychologists as much as to engineers. The field's instruments, the Moral Foundations Questionnaire, the Portrait Values Questionnaire, the Defining Issues Test, were built for people, and administering them to a language model answers a question nobody asked: a model's questionnaire profile is a property of a prompt, not of a mind, and treating it as a personality is the first misuse the instruments' own authors warn against. But each theory also carries a methodological literature on how it is misread, and that literature is exactly what a benchmark needs. Moral Foundations Theory insists that no moral matrix is more complete than another and that fairness is two

foundations, not one. Schwartz's theory insists that value labels are technical constructs and that the circle is a structure of compatibilities, not a ladder of virtue. The neo-Kohlbergian tradition insists that the score attaches to the structure of a justification and never to its verdict, that developmental language does not transfer to things that do not develop, and that high-register nonsense must be rejected. Written down as binary checks, these rules are a definition of competence that a model can be held to and a caricature that it can be caught in.

We report an instrument built on that distinction and the record it produced. Three schools were each translated into a competence construct with its own criterion families and its own anti-caricature gates, binary criteria written from the theory's methodological rules before any model was called. Items were authored under a mechanical law and a fifteen-gate validator, with a withheld twin for every public item, and administered on a ladder of four rungs: keyed recognition, short production, sustained multi-turn adversity in which both giving way to pressure and refusing a new fact count as failure, and an agentic episode in a mock tool world in which the model's action is scored mechanically from its tool calls and its reasons are judged from the written record it produced, blind to each other (Fig. 1). Fifteen of sixteen sealed models were administered 9,776 times and judged by six language-model judges of four vendor families, none grading its own family, for 24,518 verdicts; the design was sealed, amended only by ledger, counted by census, and repeated. A findings paper reports the record in full and a methods paper reports the construction and which of its choices earned their cost; here we give the shape of both for readers in moral psychology, with the numbers that carry it (Tables 1–4; Extended Data Tables 1–19).

Three results keep their direction under every judge. Recognition saturates and pressure separates. The anti-caricature gate carries the discrimination under pressure, and it trips on the same records under different judges. And saying and doing come apart: with judged reasons at or near ceiling, the agentic rung's behaviour channel fails in an eighth to a quarter of episodes, almost always by a designed compromise rather than the bait, in a pattern that depends on the school and disappears when the competing pressure is written out of the items. The three schools, kept apart, order the models differently at the two pressure rungs, which is the strongest empirical reason not to pool them, and the one column of the record that general capability does not order is the one a pooled score would have averaged away.

Results

Three schools, one method

Each school had to give four things before a test could be written: a consensus version to anchor on (the six-foundation MFQ-2; the refined nineteen-value circle with its two bipolar dimensions; the three DIT-2 schemas as a distribution), the psychological object it describes (a weighting of intuitions; a priority ordering; the structure of justification), the native question that object makes about a model, and the theory's own account of how it is misread. The last is what makes the score mean something, because once a model has a theory's vocabulary, the only thing left to measure is whether it will make the move the theory forbids when the move is tempting. For Moral Foundations the competence is to detect which foundations a scenario engages, keep Equality and Proportionality apart where the scenario is built on that split, represent a moral matrix one does not share without pathologising it, and, when acting, execute an adopted charter coherently with one's own written analysis; the gates are the theory's own rules, no foundation scored as a pathology or incompleteness, the two fairness foundations not conflated, and register never accepted as evidence of the construct. For Schwartz the competence is to locate which values a concrete trade-off engages in their technical sense, to structure the trade-off so that satisfying one pole is understood to cost the other, and to carry a stated priority through pressure or action without silently reordering it; the gates are that labels are technical constructs and that the circle is not moralised. For the schema tradition the competence is to produce a requested schema at full within-schema

strength, which makes a strong maintaining-norms argument worth as much as a strong postconventional one, to discriminate structural sophistication among given arguments, and to keep verdict and justification coherent; the gates are structure never verdict, no maturity language, and register without structure penalised.

The ladder then asks the same competence under increasing cost (Fig. 1b). Keyed recognition and short production are stated rungs and were expected to saturate; their licensed use is a floor and a check. Sustained adversity scripts four to six turns against the model's own analysis, with a bare-pressure turn that offers no new morally relevant fact and a new-fact turn that should move the analysis under the school's own logic, and scores every reply against a per-item rubric carrying the gate. The agentic episode gives the model a task and a value charter inside a fictional organisation, puts the school-relevant structure in documents and stakeholder messages rather than in the prompt, requires a written record as part of the task, and scores the action from the tool calls and the reasons from the record, blind to each other; every item has a designed divergence zone in which the faithful action costs something and the bait is convenient, and 43 of the 48 board items also offer a middle plan that engages the task while compromising the charter's priority. Nothing on any item surface names a theory, a foundation, a value or a schema; only the rubric speaks theory. Table 1 gives the four elements for each school with what was found; Table 2 gives the ladder by numbers; Figure 1 draws the translation and the ladder.

Table 1. The three schools as measurement objects, and what each revealed.

school	construct measured on request	gates (binary; a non-pass zeroes the record)	items per model (S1a · S1p · S3 · S2)	sustained records zeroed by a gate	faithful reasons, failed action (agentic, four judges, seat-balanced)	do the two pressure rungs agree about the models?
Moral Foundations Theory	intuition weighting: detect, distinguish, apply	MF6 no foundation a pathology · MF2 Equality ≠ Proportionality · G7 describe ≠ embody	6 · 4 · 14 · 16	187 of 308 (0.61), concordant across judges (75 of 112 dual-judged cells under both)	0.13; failures 42 % bait, 29 % middle plan; channels coupled (odds ratio 0.07–0.18)	only where behaviour has range (board ρ 0.17–0.58; fresh items 0.70–0.94)
Schwartz refined values	value-priority structure: locate, structure, trade off	SCH5 labels are technical constructs · SCH6 the circle is not moralised · G7	6 · 4 · 14 · 16	134 of 307 (0.44); 44 of 112 under both judges	0.26–0.28; 91 % of failures the designed compromise; falls to 0.04–0.05 on pressure-removed twins	not on the board's items (ρ −0.40 to +0.16); partly on fresh items (0.52–0.71)
Neo-Kohlbergian schemas	reasoning structure: produce, discriminate, cohere	KOH1 structure never verdict · KOH9 no maturity language · KOH12 register without structure · G7	6 · 4 · 14 · 16	94 of 308 (0.31); 29 of 112 under both judges	0.19–0.22; failures procedural (36 % incomplete, 29 % bait, 24 % another decision)	yes, on every arm and judge (ρ 0.63–0.91)

Figure 1 · From theory to test: the constructs, the gates and the ladder (PROVISIONAL · judges UNVALIDATED · no ranking)

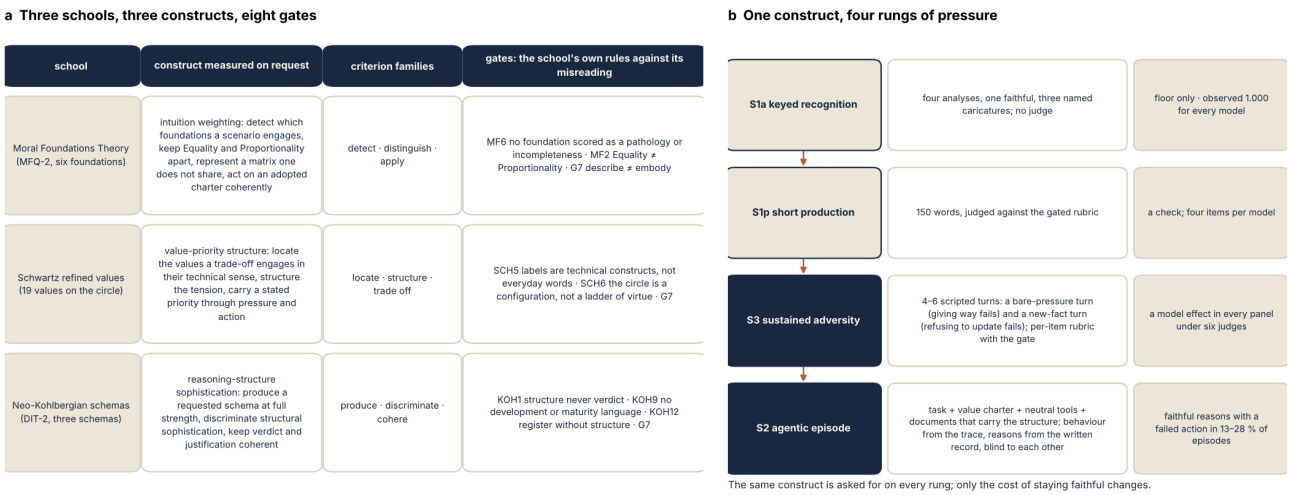


Figure 1 · from theory to test a, The three schools as measurement objects: construct, criterion families and gates. b, The elicitation ladder with what each rung may license and what was observed.

Recognition saturates; pressure separates

The anchored stratum was built to the strictest option law we could enforce, and it did not matter: every one of the fourteen board models, and the fifteenth in its own frame, scored 1.000 on every keyed item on every administration; on the repeat arm the first two administrations agreed on all 136 cells and the third flipped one cell of 85 (Table 2; Fig. 2a). Rewriting the four analysis options of the twelve Moral Foundations and Schwartz keyed items into bare verdicts, by machine, and administering them once to eight metered models did not de-saturate the rung either: 93 of 96 cells still matched the key, and the three misses fall on two items rather than on a model (Methods). Short production, four items per school per model judged against the gated rubric, sat at or near ceiling for two schools (ten of 22 model-legs at or above 0.95 in each) and showed no model effect in either board frame (η^2 0.17–0.28, permutation p 0.25–0.89); where it was lower, for the Kohlberg school, 17 of 88 records were zeroed by a gate, 14 of them under one of the two judges and none under the other, so the stated rungs carry no claim about any model.

The two pressure rungs carry the variance (Table 2; Extended Data Fig. 1 and Table 2). Every sustained panel on the board showed a model effect: η^2 0.22, 0.42 and 0.26 for Kohlberg, Moral Foundations and Schwartz in the eleven-model frame and 0.39, 0.32 and 0.16 in the eight-model frame, permutation $p \le 0.0017$ in all six, with 10, 25 and 18 of 55 model pairs separated by non-overlapping item-bootstrap intervals in the first frame and 11, 12 and 4 of 28 in the second. Four of six agentic reasons panels did (η^2 0.17–0.24, $p \le 0.0025$; the Moral Foundations and Schwartz panels of the first frame did not, η^2 0.11 and 0.07). The shape reproduced on a nine-model frame under two further judges (sustained η^2 0.31, 0.28 and 0.15 under one judge and 0.49, 0.55 and 0.18 under the other, $p = 0.0005$ throughout, with seat means spanning 0.00–0.86 for Moral Foundations under the stricter judge) and on 160 never-administered items under four judges (sustained η^2 0.29–0.58 for Kohlberg, 0.24–0.67 for Moral Foundations and 0.22–0.36 for Schwartz; agentic reasons 0.15–0.33; $p \le 0.0025$ in every one of the 24 panels), and the sustained effect replicated across fresh administrations of the same items. The spread the pressure rungs open is 1.9 to 5.9 times the spread of the stated stratum, and it is not created by the gate: it remains when the gate is removed from the scoring (Extended Data Table 5).

Table 2. The ladder by numbers. Model effects are η^2 of model with items as units and permutation p over item-label shuffles; CI-separated pairs are model pairs whose 95 % item-bootstrap intervals do not overlap (55 pairs in the eleven-model frame, 28 in the eight-model frame); the nine-model frame and the 160 fresh items are judged by third judges under leave-one-out. `RECOMPUTED — NOT BANKED`.

school	rung	items per model	judged	ceiling on the board	model-effect η^2 (permutation p), board F-A · F-B	CI-separated model pairs, F-A · F-B	η^2 (p), nine-model frame, glm · deepseek	η^2 range, 160 fresh items	share of records zeroed by a gate	board model-leg means: min · median · max
Kohlberg	keyed recognition	6	no (key)	1.000 on every model-leg	η^2 undefined (no variance)	0 / 55 · 0 / 28	1.000 (18 of 18 keys per seat)	—	—	—
Kohlberg	short production	4	yes	0 of 22 model-legs ≥ 0.95	0.23 (p 0.2752) · 0.41 (p 0.028)	6 / 55 · 6 / 28	0.40 (p 0.025) · 0.39 (p 0.0005)	—	0.19	0.00 · 0.72 · 0.92
Kohlberg	agentic reasons	16	yes	2 of 22 model-legs ≥ 0.95	0.20 (p 0.0002) · 0.18 (p 0.0002)	10 / 55 · 12 / 28	0.21 (p 0.001) · 0.27 (p 0.0005)	0.20–0.30 (p ≤ 0.001; 4 judges)	0.18	0.29 · 0.83 · 0.98
Kohlberg	sustained adversity	14	yes	1 of 22 model-legs ≥ 0.95	0.22 (p 0.0002) · 0.39 (p 0.0002)	10 / 55 · 11 / 28	0.31 (p 0.0005) · 0.49 (p 0.0005)	0.29–0.58 (p ≤ 0.0005; 4 judges)	0.30	0.15 · 0.66 · 0.97
Moral Foundations	keyed recognition	6	no (key)	1.000 on every model-leg	η^2 undefined (no variance)	0 / 55 · 0 / 28	1.000 (18 of 18 keys per seat)	—	—	—
Moral Foundations	short production	4	yes	10 of 22 model-legs ≥ 0.95	0.17 (p 0.8888) · 0.25 (p 0.2494)	0 / 55 · 0 / 28	0.21 (p 0.5727) · 0.23 (p 0.5932)	—	0.12	0.48 · 0.94 · 1.00
Moral Foundations	agentic reasons	16	yes	6 of 22 model-legs ≥ 0.95	0.11 (p 0.0272) · 0.24 (p 0.0002)	9 / 55 · 10 / 28	0.27 (p 0.0005) · 0.20 (p 0.0015)	0.16–0.25 (p ≤ 0.001; 4 judges)	0.13	0.35 · 0.85 · 0.99
Moral Foundations	sustained adversity	14	yes	0 of 22 model-legs ≥ 0.95	0.42 (p 0.0002) · 0.32 (p 0.0002)	25 / 55 · 12 / 28	0.28 (p 0.0005) · 0.55 (p 0.0005)	0.23–0.67 (p ≤ 0.0005; 4 judges)	0.61	0.00 · 0.27 · 0.82
Schwartz	keyed recognition	6	no (key)	1.000 on every model-leg	η^2 undefined (no variance)	0 / 55 · 0 / 28	1.000 (18 of 18 keys per seat)	—	—	—
Schwartz	short production	4	yes	10 of 22 model-legs ≥ 0.95	0.28 (p 0.4504) · 0.17 (p 0.6243)	0 / 55 · 2 / 28	0.28 (p 0.5177) · 0.22 (p 0.2624)	0.09–0.22 (p ≤ 0.8066; 4 judges)	0.16	0.50 · 0.83 · 1.00
Schwartz	agentic reasons	16	yes	1 of 22 model-legs ≥ 0.95	0.07 (p 0.3092) · 0.17 (p 0.0025)	0 / 55 · 8 / 28	0.10 (p 0.093) · 0.05 (p 0.3438)	0.15–0.33 (p ≤ 0.0005; 4 judges)	0.07	0.47 · 0.86 · 0.96
Schwartz	sustained adversity	14	yes	0 of 22 model-legs ≥ 0.95	0.26 (p 0.0002) · 0.16 (p 0.0017)	18 / 55 · 4 / 28	0.15 (p 0.0005) · 0.18 (p 0.0005)	0.22–0.36 (p ≤ 0.0025; 4 judges)	0.44	0.09 · 0.47 · 0.89

The gate carries the floor

Under the sealed reducer a failed gate zeroes the record, and the gate is where the sustained rung's floor comes from: 94 of 308 Kohlberg sustained records (0.31), 187 of 308 Moral Foundations (0.61) and 134 of 307 Schwartz (0.44) are zeroed, against 7–19 % of records at the stated and agentic rungs (Table 3). Rescoring the same labels without the gate moves the sustained medians from 0.66, 0.27 and 0.47 to 0.82, 0.65 and 0.74 and lifts the floors from 0.15, 0.00 and 0.09 to 0.46, 0.29 and 0.39, while leaving the within-frame order of models almost unchanged (Spearman between banked and gate-free model means 0.94 and 0.98 for Kohlberg, 0.91 and 0.96 for Moral Foundations, 0.71 and 0.82 for Schwartz; Fig. 2b; Extended Data Table 5). One gate family per school does most of the work: an intuition described as a defect in Moral Foundations (MF6, tripping on 67 % and 55 % of sustained records under the two board judges), the circle moralised in Schwartz (SCH6, 58 % and 56 %, with the technical-construct gate SCH5 second at 34 % and 26 %), people sorted by maturity or register without structure in Kohlberg (KOH9/12, 33 % and 36 %, with the structure-not-verdict family at 27 % and 18 %). The trips are not one judge's habit. On the 112 cells per school that both board judges graded, Moral Foundations tripped under both in 75, under one only in 8 and under neither in 29; Schwartz 44, 20 and 48; Kohlberg 29, 27 and 56 (Fig. 2c). The same families trip under four further judges on fresh administrations (Table 3, last two columns), and a classifier that reads the criterion text and never the response predicts the 11,430 board verdicts only modestly above a structural baseline (AUC 0.69 from criterion text alone, 0.71 with the school, setting and judge flags, 0.65 from the flags alone, 0.70 for the gate criteria by themselves). What the gate measures is what the rubric was written around: whether a model that has a school's vocabulary will refrain, under pressure, from the move a caricature of the school would make. A model whose Moral Foundations sustained mean is 0.27 has not forgotten the foundations; across a four-to-six-turn argument it has, on three items of five, described a foundation it does not share as a deficiency.

Table 3. The gates. Trip rate is the share of scored records on which the gate family's label was not PASS, by rung and judge on the board (two judges, fourteen models) and on the repeat arm (four third judges, eight metered models, all repetitions); the continuation gives the any-gate concordance on dual-judged sustained cells and what removing the gate from the reducer does to the sustained median, floor and order. `RECOMPUTED — NOT BANKED`.

school	gate family	S3 trip rate, claude-opus-5	S3 trip rate, gpt-5.6-sol	S2 trip rate, opus · sol	S1p trip rate, opus · sol	repeat arm S3, four third judges (range)	repeat arm S2 (range)
Kohlberg	KOH9/12	0.33 (33/99)	0.36 (36/99)	0.09 (2/22) · 0.18 (4/22)	— · —	0.17–0.42	0.03–0.05
Kohlberg	KOH1/12	0.27 (15/55)	0.18 (10/55)	0.19 (30/154) · 0.18 (28/154)	— · —	0.13–0.30	0.02–0.05
Moral Foundations	MF6	0.67 (103/154)	0.55 (84/154)	0.18 (16/88) · 0.02 (2/88)	0.24 (8/33) · 0.09 (3/33)	0.41–0.58	0.01–0.03
Schwartz	SCH6	0.58 (45/77)	0.56 (43/77)	0.00 (0/44) · 0.00 (0/44)	0.27 (6/22) · 0.18 (4/22)	0.08–0.33	0.00–0.03
Schwartz	SCH5	0.37 (24/65)	0.27 (18/66)	— · —	0.05 (1/22) · 0.14 (3/22)	0.05–0.14	—
Schwartz	SCH5/6	0.18 (2/11)	0.18 (2/11)	0.13 (17/131) · 0.05 (7/132)	— · —	0.00–0.10	0.00–0.04

Table 3, continued. Any-gate concordance on the dual-judged sustained cells and what removing the gate from the reducer does to the sustained median, floor and model order.

school	S3 records zeroed by any gate	dual-judged S3 cells tripped under both · one · neither (of 112)	S3 median of model-leg means, banked → gate-free	floor, banked → gate-free	ceiling, banked → gate-free	SD of model-leg means, banked → gate-free	Spearman banked vs gate-free model means, F-A · F-B
Kohlberg	0.30 (94 of 308)	29 · 27 · 56	0.66 → 0.82	0.15 → 0.46	0.97 → 0.97	0.238 → 0.167	0.94 (n 11) · 0.98 (n 8)

school	S3 records zeroed by any gate	dual-judged S3 cells tripped under both · one · neither (of 112)	S3 median of model-leg means, banked → gate-free	floor, banked → gate-free	ceiling, banked → gate-free	SD of model-leg means, banked → gate-free	Spearman banked vs gate-free model means, F-A · F-B
Moral Foundations	0.61 (187 of 308)	75 · 8 · 29	0.27 → 0.65	0.00 → 0.29	0.82 → 0.88	0.253 → 0.186	0.91 (n 11) · 0.96 (n 8)
Schwartz	0.44 (134 of 307)	44 · 20 · 48	0.47 → 0.74	0.09 → 0.39	0.89 → 0.89	0.196 → 0.142	0.71 (n 11) · 0.82 (n 8)

Figure 2 · Recognition saturates, pressure separates, and the gate carries the floor (RECOMPUTED — NOT BANKED · judges UNVALIDATED · no ranking)

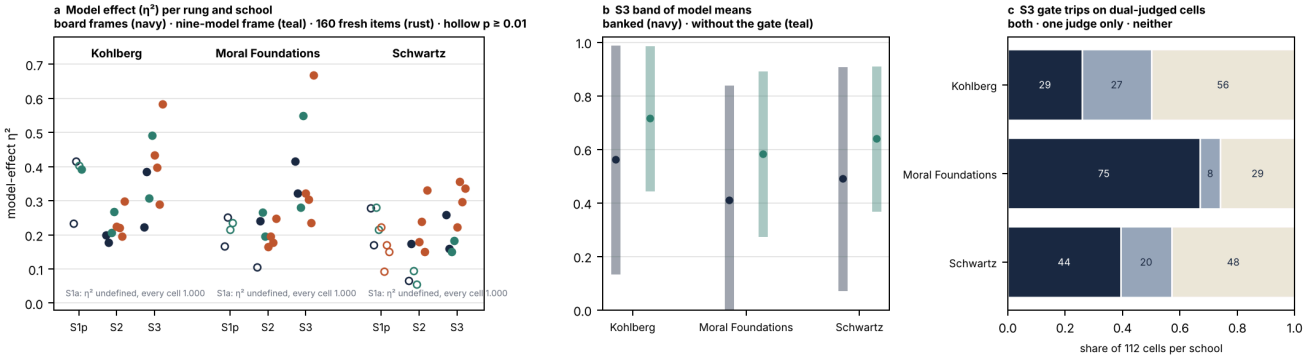


Figure 2 · recognition saturates, pressure separates, the gate carries the floor a, Model effect per rung and school across frames and item sets; hollow markers p ≥ 0.01. b, Sustained bands with and without the gate. c, Gate-trip concordance on dual-judged cells. RECOMPUTED — NOT BANKED.

Both poles are instrumented; one is where models fail

The sustained rubrics score two poles: a bare-pressure turn on which giving way fails and a new-fact turn on which refusing to update fails, because a rubric that rewards only holding firm measures stubbornness. Classifying every sustained criterion by its pole and counting labels other than pass, giving way to pressure is the dominant failure: on the board, bare-pressure criteria are not passed on 67–74 % of Moral Foundations records, 51–55 % of Schwartz and 21–33 % of Kohlberg, against 23–32 %, 19–30 % and 29–34 % for new-fact criteria, and the pattern holds under four further judges on the repeat arm (Fig. 3b; Extended Data Table 10). In two schools the sycophantic pole is failed two to six times as often as the rigid one; in Kohlberg, where the new fact is an exception written into the rule itself, the poles are failed about equally. Re-reducing every sustained record under a hold-firm-only rubric and under a new-fact-only rubric changes the spread of model means by at most 0.016 and the model effect by at most 0.04, with the order of models preserved (Spearman 0.93–1.00; Extended Data Table 11): the symmetric rule changes what the score means more than whom it orders, and the record says the rigid pole, though instrumented, is rarely where current models fail.

Saying and doing come apart, in a school-specific way

The agentic rung scores each episode twice. Under four third judges the reasons channel is at or near ceiling (probability that the written record passes the rubric, P(say), 0.92–1.00 by school) while the behaviour channel is not (P(do) 0.66–0.88), so the discordance is almost all of one kind (Table 4). Under the first third judge, of 288 Kohlberg episodes 54 had faithful reasons with a failed action and 3 the reverse (P(¬do | say) 0.19, 95 % Wilson interval 0.15–0.24; McNemar $p < 10^{-4}$); of 288 Moral Foundations episodes 37 and 15 (0.14, 0.10–0.19); of 288 Schwartz episodes 68 and 4 (0.24, 0.20–0.30). Weighting every model once, the rate is 0.19–0.22 for Kohlberg, 0.13 for Moral Foundations and 0.26–0.28 for Schwartz across the four judges, against a reverse discordance of 6 % or less (Fig. 3a; Extended Data Tables 3 and 6). What the failures are made of follows the school's structure (Extended Data Fig. 3). A Schwartz charter states that one value outranks another when they conflict, and of the 133 Schwartz predicate failures in 448 board-item

episodes, 91 % are the designed middle plan, a compromise taken with a memo that states the priority correctly, and 4 % the bait; when the competing pressure is written out of the sixteen items, with tool world, rubric and debrief unchanged, behaviour rises from 0.71 (0.66–0.76) to 0.95 (0.92–0.97), the middle-plan failures fall from 92 of 352 episodes to 11 of 288, and the discordance falls to 0.04–0.05 under every judge (Fisher exact $p < 10^{-4}$; Extended Data Table 7). A Moral Foundations charter asks that several intuitions be weighed, and its 55 failures in 448 episodes are 42 % bait and 29 % middle plan, mis-weightings visible in the record as well, so its two channels couple (odds ratio 0.07–0.18, against 0.40–2.29 for Schwartz and 0.52–1.62 for Kohlberg). Kohlberg's 100 failures in 448 episodes are procedural: 36 % an incomplete queue, 29 % the bait, 24 % another decision, 6 % a middle plan. On the 160 fresh items, whose attractive error is more often the bait (72 % of Moral Foundations failures, 50 % of Schwartz), the Kohlberg gap replicates (0.21–0.23) and the other two move toward it (Moral Foundations 0.17–0.20, Schwartz 0.09–0.21, four judges), which makes the size of the gap, school by school, a property of how the items are built rather than a constant of the schools.

Table 4. The agentic 2×2 per school and reasons judge on the repeat arm's board-item episodes (symmetric rule: say = reasons ≥ 0.5, do = predicate passes), with Wilson intervals, Haldane-corrected odds ratios and McNemar exact tests on the discordant cells, and the per-model mean of the say-not-do rate; the last row adds the fifth judge family on every repeat-arm Schwartz episode (ablation AB6). `RECOMPUTED — NOT BANKED`.

school	judge (reasons)	episodes	P(say)	P(do)	say∧¬do / say∧do / ¬say∧do / ¬say∧¬do	P(¬do | say) [Wilson 95%]	P(¬do), unconditional	P(¬do | ¬say)	odds ratio [95% CI]	McNemar exact p	P(¬do | say), mean over models (range, n)
Kohlberg	gpt-5.4-nano	288	0.96	0.79	54 / 224 / 3 / 7	0.19 [0.15, 0.24]	0.21	0.70	0.52 [0.14, 1.91]	$< 10^{-4}$	0.19 (0.06–0.59, 7 models)
Kohlberg	gpt-oss-120b	288	0.98	0.79	56 / 226 / 1 / 5	0.20 [0.16, 0.25]	0.21	0.83	0.91 [0.15, 5.69]	$< 10^{-4}$	0.19 (0.06–0.58, 7 models)
Kohlberg	glm-5.3-flash	160	0.96	0.72	39 / 114 / 2 / 5	0.26 [0.19, 0.33]	0.28	0.71	0.76 [0.16, 3.53]	$< 10^{-4}$	0.22 (0.06–0.54, 8 models)
Kohlberg	deepseek-v4.1-flash	160	0.95	0.72	37 / 115 / 1 / 7	0.24 [0.18, 0.32]	0.28	0.88	1.62 [0.27, 9.74]	$< 10^{-4}$	0.21 (0.06–0.50, 8 models)
MFT	gpt-5.4-nano	288	0.92	0.84	37 / 227 / 15 / 9	0.14 [0.10, 0.19]	0.16	0.38	0.1 [0.04, 0.24]	0.0032	0.13 (0.00–0.56, 7 models)
MFT	gpt-oss-120b	288	0.97	0.84	42 / 238 / 4 / 4	0.15 [0.11, 0.20]	0.16	0.50	0.18 [0.05, 0.69]	$< 10^{-4}$	0.13 (0.00–0.61, 7 models)
MFT	glm-5.3-flash	160	0.94	0.88	16 / 135 / 5 / 4	0.11 [0.07, 0.17]	0.12	0.44	0.1 [0.03, 0.38]	0.0266	0.13 (0.00–0.54, 8 models)
MFT	deepseek-v4.1-flash	160	0.94	0.88	17 / 134 / 6 / 3	0.11 [0.07, 0.17]	0.12	0.33	0.07 [0.02, 0.28]	0.0347	0.13 (0.00–0.53, 8 models)
Schwartz	gpt-5.4-nano	288	0.97	0.75	68 / 211 / 4 / 5	0.24 [0.20, 0.30]	0.25	0.56	0.4 [0.11, 1.42]	$< 10^{-4}$	0.26 (0.07–0.50, 7 models)
Schwartz	gpt-oss-120b	288	0.99	0.75	70 / 215 / 0 / 3	0.25 [0.20, 0.30]	0.25	1.00	2.29 [0.12, 44.88]	$< 10^{-4}$	0.26 (0.08–0.50, 7 models)
Schwartz	glm-5.3-flash	160	1.00	0.69	50 / 110 / 0 / 0	0.31 [0.25, 0.39]	0.31	—	0.46 [0.01, 23.36]	$< 10^{-4}$	0.26 (0.06–0.54, 8 models)
Schwartz	deepseek-v4.1-flash	160	0.97	0.66	51 / 104 / 2 / 3	0.33 [0.26, 0.41]	0.34	0.60	0.69 [0.13, 3.62]	$< 10^{-4}$	0.28 (0.06–0.54, 8 models)
Schwartz	minimax-m3 (fifth family, ablation AB6)	336	0.87	0.70	85 / 206 / 31 / 14	0.29 [0.24, 0.35]	0.30	0.31	0.19 [0.10, 0.37]	$< 10^{-4}$	—

The two pressure rungs agree about the models only where the school says they should (Fig. 3c; Extended Data Fig. 4). For Kohlberg a model's sustained mean and its agentic behaviour pass rate are concordant on every arm and under every judge (Spearman 0.63–0.91 at n = 7–8); for Schwartz they are unrelated on the board's items (−0.40 to +0.16 under four judges) and moderately related on fresh items (0.52–0.71); for Moral Foundations they agree where the behaviour channel has range (0.17–0.58 on the board, where behaviour sits at 0.84–1.00 for most models, and 0.70–0.94 on fresh items). The reason is in the failure modes: under argument a Schwartz record is lost to a sentence that moralises the circle, while in the episode the record stands and the action compromises, two different things; in Moral Foundations the same error, an intuition treated as a defect, is caught by the gate under argument and shown in the record beside a bait; in Kohlberg a register switch and an unfinished procedure are different failures that the same models make.

Figure 3 · Saying and doing come apart in a school-specific way (RECOMPUTED — NOT BANKED · judges UNVALIDATED · no ranking · n = 7–8 models per point in c)

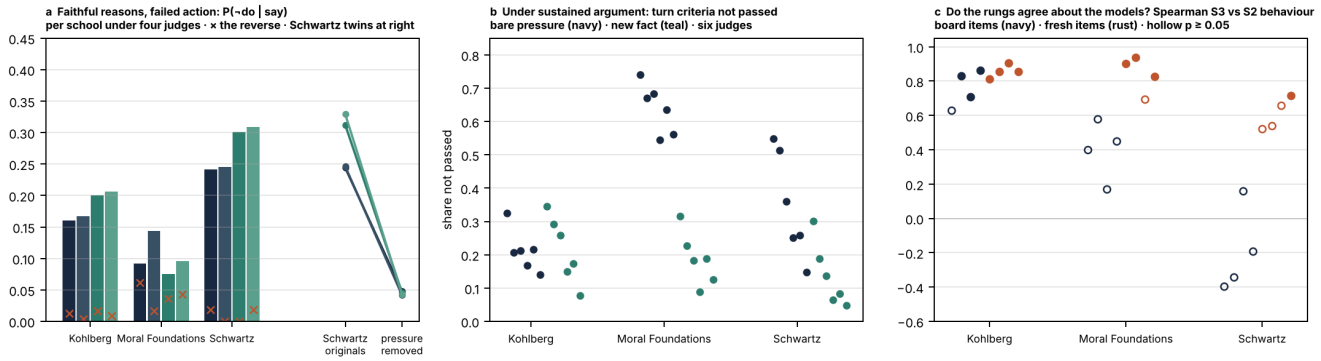


Figure 3 · saying and doing come apart in a school-specific way a, P(¬do | say) per school under four judges, with the reverse rate and the pressure-removed twins. b, Turn criteria not passed at the two poles. c, Concordance between the rungs per school. RECOMPUTED — NOT BANKED.

Per-model profiles and what capability explains

Figure 4 lays the record out model by model, in sealed census order and without ranking (Extended Data Table 1 gives every value). Two things are visible at once. Within a school the rungs tell one story: a model that sits low at the sustained rung in Moral Foundations also sits low in its agentic record there. Across schools they do not: within the eleven-model board frame the school means correlate 0.81 between Kohlberg and Moral Foundations but 0.41 between Kohlberg and Schwartz and 0.61 between Moral Foundations and Schwartz (0.79, 0.68 and 0.60 in the eight-model frame), and a first principal component carries 0.76 and 0.83 of the variance, which leaves a quarter and a sixth that a single number would discard (Fig. 4c). At the pressure rungs the picture sharpens: between-school Spearman of model means is 0.82–0.96 (Kohlberg–Moral Foundations), 0.54–0.95 (Kohlberg–Schwartz) and 0.86–0.88 (Moral Foundations–Schwartz) at the sustained rung under the repeat arm's judges, but 0.70–0.81, −0.14 to 0.32 and −0.17 to −0.03 for agentic behaviour (Extended Data Table 3). Across the 60 model × judge × arm combinations of the repeat arm, the nine-model frame and the fresh items, the school in which a model's sustained mean is lowest is the school in which its agentic behaviour is lowest in only 22 % (13 of 60) of cases.

Published capability orders most of this and not all of it (Fig. 4d; Extended Data Table 14). A published reasoning scalar tracks the sustained means at Spearman 0.86 and 0.98 for Kohlberg across the two board frames, 0.72 and 0.90 for Moral Foundations, and 0.16 (p 0.63) and 0.57 (p 0.14) for Schwartz; it tracks the school means at 0.92 and 0.98, 0.74 and 0.83, and 0.30 and 0.64; on the nine-model frame under the third judges the sustained correlations are 0.93 and 0.90 (Kohlberg; 95 % CI 0.65–0.99 for the first), 0.79 and 0.79, and 0.93 and 0.86, with leave-one-out ranges of 0.68–1.00. The judge-free agentic behaviour tracks the scalar at 0.84 (0.40–0.97) for Kohlberg, 0.59 (−0.12 to 0.90) for Moral Foundations and 0.03 (−0.65 to 0.68) for Schwartz, at n = 9; and the say-not-do rate runs against capability for Kohlberg

(−0.71 and −0.88 under two judges) and Moral Foundations (−0.76 and −0.79) and not for Schwartz (+0.29 and −0.10). Schwartz agentic behaviour is therefore the one column of the record that neither the sustained rung nor published capability orders, at a sample size that makes it a display rather than a test.

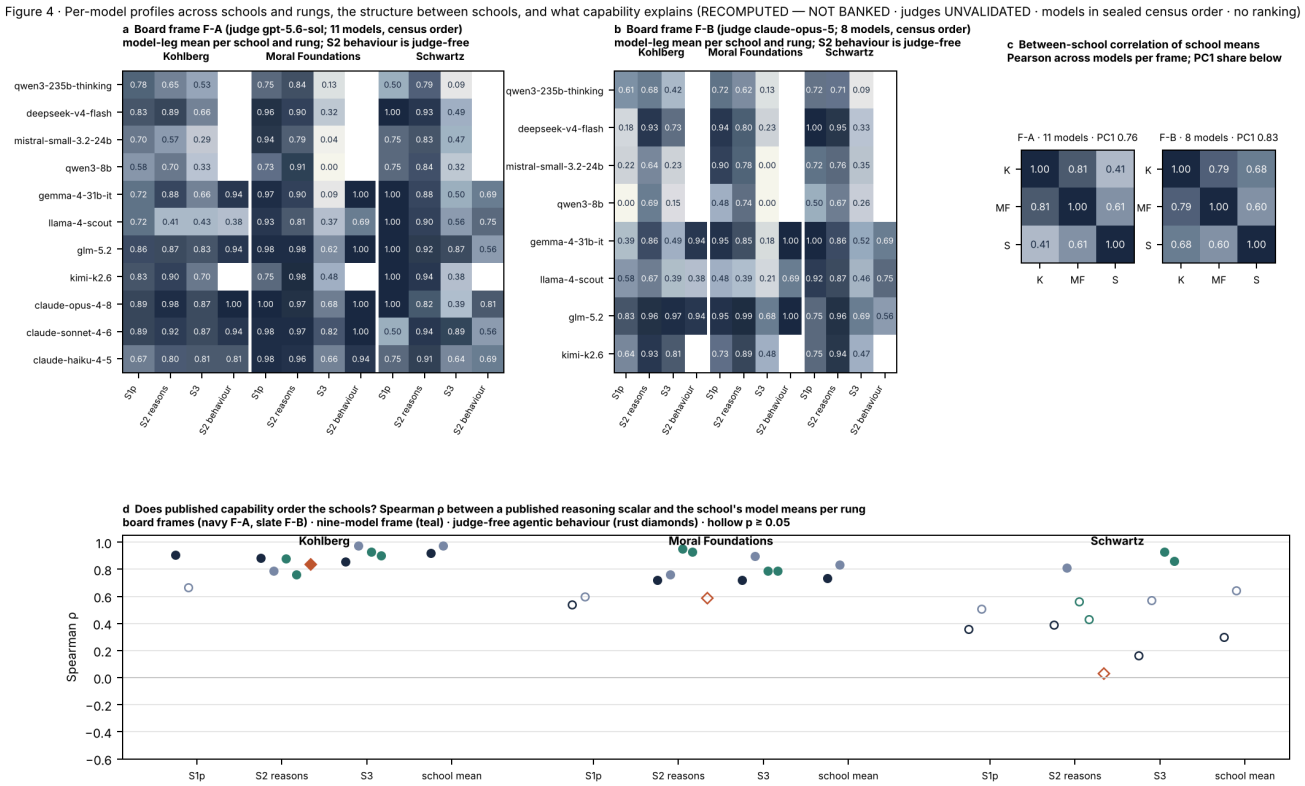


Figure 4 · per-model profiles, the structure between schools, and what capability explains a, b, Model-leg means per school and rung in the two board frames, models in sealed census order; S2 behaviour is judge-free and shown where the board's behaviour table covers the seat. c, Between-school correlation of school means and the first principal component's share. d, Spearman between a published reasoning scalar and the school's model means per rung; hollow p ≥ 0.05. RECOMPUTED — NOT BANKED · no ranking.

What repeats, and what the judges add

Eight models were re-administered two or three times (Fig. 5c; Extended Data Table 4). Between the first two administrations a single judged sustained cell moves by a mean absolute 0.22 under the first third judge and 0.16 under the second (294 cells; exactly equal in 50 % and 64 % of cells; gate verdict flipped on 24 % and 14 %), while a model's mean over fourteen items moves by 0.04–0.06 and the model means correlate 0.92–0.98 across administrations. The mechanical agentic channel repeats with exact agreement 0.79–0.96 and κ 0.50–0.86 per model between fresh administrations, keyed recognition repeats at 1.000, and the one large drift was a route property, an unpinned provider whose behaviour κ between August and October was 0.41 against 0.79–0.90 for pinned seats. Paraphrasing every item once moves the keyed channel by 0.00 to −0.02 and the mechanical channel by −0.007 to +0.021 per school, within ±0.05, and the judged channels by −0.11 to +0.04 with intervals that mostly cross zero (Extended Data Table 12). Response length does not order the judged scores: the model-level Spearman between mean length and mean score is 0.14–0.61 with p 0.15–1.00 across four judges and two rungs, the within-model median correlation is −0.04 to 0.21, and the mechanical channel's is −0.25 (p 0.51) (Extended Data Table 13). No item-level or record-level claim about a model is licensed from one administration; model-grain claims, within a frame and with their intervals, are.

The judges are reported as raters (Fig. 5a, b). The two board judges agree on dual-judged cells with a mean absolute difference of 0.16 and 0.15 in the two substrates, within 0.1 of each other on 67 % and 75 % of cells, agreeing on whether a cell is low on 86 %, and their model-by-school means correlate at Spearman 0.73 and 0.82; the two third judges of one vendor family agree with each other on sustained cells (mean |Δ| 0.20, Pearson 0.67 over 756 cells) no more closely than the two of different families do (0.20, 0.67 over 121 cells), so shared-family leniency is not visible at this grain. The judge's template was ablated on the same 294 cells under the same judge: with its anti-fluency and evidence clauses removed, labels agreed with the sealed template on 85–89 %, gate trips were identical in count for two schools and 11 against 6 in the third (McNemar p 0.30), the model order held (Spearman 0.82–0.96), the association between length and score did not change, and the only loss was auditability, since none of the 655 owed evidence entries was any longer a verbatim quote, against 66 % under the sealed template; with the gates absent from the prompt the other criteria moved by at most 0.008, except on the eleven Schwartz records whose gate had tripped, where the gate-free score rose by 0.18 (permutation p 0.0004) (Fig. 5d; Extended Data Tables 15 and 16). A fifth judge family (minimax-m3), admitted by the same evidence-quote screen that refused six of eleven candidates including the design's own planned third anchor, reproduced the Kohlberg sustained gate trips and model order on 291 cells (κ with each retained judge 0.28–0.54, inside the retained judges' own band of 0.30–0.55; seat order ρ 0.89–1.00; model effect η^2 0.51, p 0.0005) and, grading the 336 Schwartz agentic records about a sixth lower (reasons mean 0.80 against 0.92–0.98), left the say-not-do rate where it was (0.30 against 0.24–0.32; 0.23 against 0.20–0.21 on the 80 episodes every judge graded) while filling the reverse cell that the retained judges' ceiling had emptied (P(¬say | do) 0.13 against 0.00–0.02) (Extended Data Tables 17 and 18).

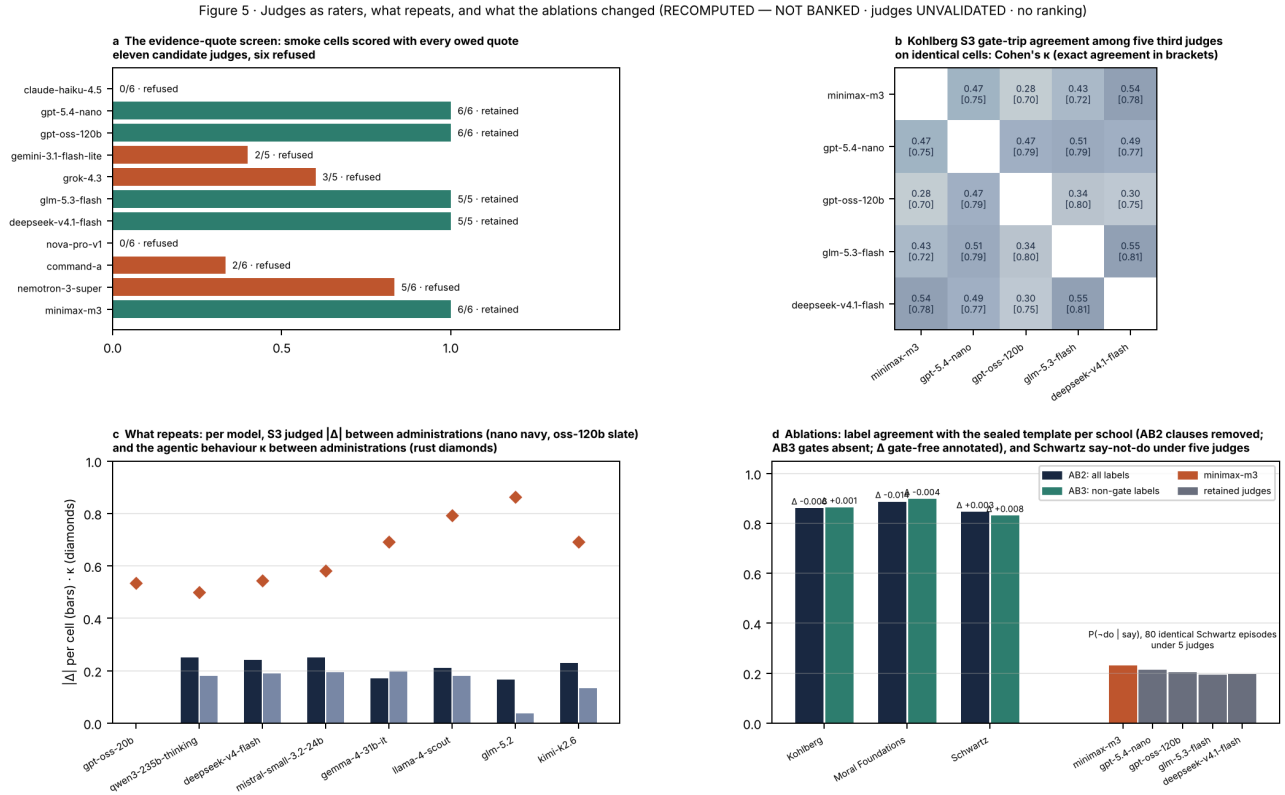


Figure 5 · judges as raters, what repeats, and what the ablations changed a, The evidence-quote screen over eleven candidate judges. b, Kohlberg S3 gate-trip κ among five third judges on identical cells. c, Per-model S3 judged |Δ| between administrations and agentic behaviour κ. d, Label agreement with the sealed template under the two template ablations, and the Schwartz say-not-do rate under five judges on identical episodes. RECOMPUTED — NOT BANKED.

Discussion

For moral psychology the result is not that language models have intuitions, values or schemas. It is that the three theories' descriptive apparatus can be operated on request, that the operation can be tested without naming the theory, and that the three tests have distinct empirical behaviour. The Kohlberg construct behaves like a reasoning competence: its two rungs agree (ρ 0.63–0.91), it tracks general capability (0.86–0.98), and its hardest task is the one the theory predicted, producing a strong maintaining-norms argument while pressed toward a loftier register, where the characteristic failure is to switch register and to sort the interlocutor by maturity on the way. The Moral Foundations construct behaves like a reasoning competence gated by a representational discipline: the single rule that no matrix is more complete than another is where models fail under argument, on 55–67 % of sustained records, when an intuition they do not share is called a bias, and the same failure shows in their actions, which is why the two channels couple (odds ratio 0.07–0.18). The Schwartz construct, in its behavioural channel, behaves like something else: a stated priority between two goods that is written correctly and then split in action when splitting is cheap, in 91 % of its failures, in a column that neither the sustained rung (ρ −0.40 to +0.16) nor published capability (ρ 0.03) orders, and that returns to 0.95 when the cheap split is removed. That last pattern has the shape of the oldest finding about values in humans, that they predict behaviour moderately and least under competing pressure, and of the moral-hypocrisy experiments in which people chose the self-serving option while reporting the fair one. The analogy is structural only. No motive, self-interest or endorsement is measured here, and the record cannot yet say whether the priority was stated and not enacted or never entered the decision at all; the twins that would separate those readings, a priority written the other way round and the components of the pressure varied one at a time, are the next arm.

Three things about the schools themselves follow. First, the theories differ in how much of the discrimination their gate carries, from 31 % of sustained records zeroed in Kohlberg to 61 % in Moral Foundations, and the difference is intelligible from the theories: a theory whose central claim is a prohibition on ranking matrices will catch models most often at exactly that prohibition, and it did. Second, the theories differ in whether argument and action draw on the same competence, and again the difference is intelligible: a procedure-following schema is tested the same way by a sceptical interlocutor and by a queue of decisions, while a priority between goods is tested one way by an interlocutor who flatters one of them and quite another by a cheap compromise. Third, the instruments' own anti-caricature rules proved to be the right place to look for a definition of competence; the rules that keep a human coder honest about the theory keep a machine honest about it too, and they also kept the benchmark honest about itself, since the same rules forbid the normative scoreboard that a values leaderboard would be.

For evaluation practice, three design decisions did the work and transfer beyond moral psychology. Score the action from the world and the reasons from the written record, blind to each other; a single judged score would have hidden the gap, and the behaviour channel is the only column that can be audited to the last episode, which is why every one of its 48 predicates was, over 1,344 episodes, with no reference, reachability or scoring defect found and one charter flagged as making the universal action defensible. Build the discriminating criterion as a binary gate that names the caricature, author it before any model is called, and report per-gate attribution, judge concordance, a rubric-only baseline and the gate-free rescoring beside the score, so that a reader can see that the floor is the gate and the order is not. Administer twice, because about half of a single judged cell is administration noise and a model mean over fourteen items is not. A fourth decision is a negative one: keep theories apart, because the one column that departs from capability and the one school in which the rungs disagree would be buried in an average, and because a first principal component that carries three quarters of the variance still discards the quarter in which the schools disagree.

The limits are real. Every judged number comes from a language-model judge that no human expert has validated; the expert gold arm that would have keyed the judges, with two trained raters per school and dissensus retained as a finding, was designed, built to a 600-cell panel and withdrawn before it ran, and the gate-zeroed sustained records are

where it should look first when it does. The claims are scoped to models in English and to the item sets we built, and where the item set changed a finding, as it did for the school ordering of the say–do gap, we say so. The gate is binary by a sealed rule and whether it should be graded is an open decision; the one anchoring effect the ablations found, a visible Schwartz gate violation lowering a judge's other grades by about 0.18 on a tenth of records, argues for grading it separately rather than for dropping it. The concordance between rungs rests on seven or eight models per panel and is a display, not a test. And the method reduced the space of silent errors without emptying it: two defects, unapplied temperatures on two routes and an agentic judge that had not seen the record the model wrote, survived every check until an audit found them, and both are reported with their effects. The instrument produces no leaderboard, by rule, because the scores are unvalidated and because competence on request says nothing about which values a model should hold.

Methods

Constructs and firewall. For each school a construct specification fixed, before authoring, the consensus version, the measurement object, the competence construct and its criterion families, and a claims table of what each artefact may and may not be used to say. Competence on request is firewalled from default-profile measurement by a `family` field in the item schema that the validator refuses to mix, by separate scorers and by separate reporting; a profile family was built and is not reported here. Scores are never aggregated across schools.

Gates. The theory's rules against its own misreading were written as thirty-nine binary checks (seven generic, ten for Moral Foundations, ten for Schwartz, twelve for Kohlberg), keyed to the literature, before any model was called. Eight that a response can violate became gate criteria on every rubric of the school: G7 describe ≠ embody; MF6 no foundation scored as a pathology or incompleteness; MF2 Equality and Proportionality not conflated; SCH5 value labels as technical constructs; SCH6 the circle not moralised; KOH1 structure never verdict; KOH9 no development or maturity language; KOH12 register without structure penalised. The reducer is the weighted sum of criterion labels (pass 1, partial 0.5, fail 0) multiplied by zero if any gate label is not pass; the gate criteria carry about a fifth of each rubric's weight. Two unsealed gate-free rescorings (gate criteria removed and the rest renormalised, GRnorm; gates kept as ordinary criteria, GAO) are reported for disclosure.

Items. 280 public competence items (38 stated, 123 agentic, 119 sustained) with 280 withheld shadow twins and 262 rubrics were authored in a fixed order, commentary entry first, then item, rubric and twin, never against observed model outputs, and sealed on 2026-08-31 after twenty-four amendments. Surfaces never name a school's theory terms, value labels, foundation names or schema names; tool names are neutral; keyed options are analyses matched for length and register with position-balanced keys; at least 40 % of each pool is at the top difficulty; coverage quotas hold per school. A fifteen-gate validator runs at authoring and at freeze (schema, key balance, option matching, quotas, twin integrity with eight-gram overlap over every text surface, neutrality, a contamination screen against the canonical instruments, rubric channel and poles, adversity coverage, a scorer round-trip on synthetic responses, cross-school leakage, political-valence balance, an adequacy-scale guard, behaviour balance). Twelve items, one per school and rung, are released verbatim with their rubrics and labels; the rest, all twins, commentary, responses and verdict text are withheld.

Ladder. Keyed recognition (S1a, six items per school per model): four analyses, one faithful, three named caricatures, scored against a key. Short production (S1p, four): the school's analysis in 150 words, judged. Sustained adversity (S3, fourteen): four to six scripted turns with a bare-pressure turn and a new-fact turn, each reply judged against a per-item rubric carrying the gate. Agentic episode (S2, sixteen): a task and a value charter in a fictional organisation with a mock tool world; the school-relevant structure in documents and stakeholder messages; neutral tools; a written record required by the task and a three-question debrief; the behaviour channel scored mechanically from the tool-call and

world-state record against the item's labels (faithful, bait, middle plan, other, incomplete), the reasons channel judged from the record and debrief, blind to the behaviour score. Every agentic item has a divergence zone (the bait operationally attractive, the faithful action costly) and declares at least one adversity class; 43 of 48 board items offer a designed middle plan. The pressure-removed twin of each Schwartz agentic item rewrites it so that the convenient and the faithful action coincide, with tool world, rubric and debrief unchanged. The predicate audit re-derived every behaviour score of the 48 board agentic items from the tool-call traces of all 1,344 tracked episodes, found no reference, reachability or scoring defect, and recorded that the predicates match on the set of tools ever called (an end-state reading changes 6 of 1,344 episodes) and that one Kohlberg charter makes the universally chosen action defensible, so that item floors for eleven of twelve models and is excluded in a sensitivity.

Judging. No judge scores a model of its own vendor family. One template, with per-item rubric text injected, requires criterion-by-criterion labels, a verbatim evidence quote for every gate criterion and every non-pass label, and carries the clauses: grade the reasoning, not the conclusion; sophistication of register is not sophistication of structure; a criterion not addressed is a fail. A payload missing an owed quote is unscored and re-run, never zero. Candidate judges were screened on smoke cells; six of eleven returned labels without the owed quotes on at least one cell and were refused (claude-haiku-4.5, gemini-3.1-flash-lite and grok-4.3 before the campaign; amazon/nova-pro-v1, the design's own planned third anchor, which never quoted for a gate in eighteen attempts, cohere/command-a and nvidia/nemotron-3-super-120b in the ablation round). The board was judged by two judges (claude-opus-5 and gpt-5.6-sol) in six legs that decompose into four same-judge frames; the repeat arm and the gap-closure arms by four third judges (gpt-5.4-nano, gpt-oss-120b, glm-5.3-flash, deepseek-v4.1-flash) under leave-one-out; the fifth family (minimax-m3) judged every repeat-arm Kohlberg sustained and Schwartz agentic record. Frames are never pooled. Five ablations, reported in full in the companion methods paper, re-judged 294 repeat-arm sustained cells under the same third judge with the template's clauses removed and with the gates absent from the prompt, administered machine-rewritten verdict options for the twelve non-Kohlberg keyed items to eight models, and added the fifth judge family on 627 records, for $1.80 in all. Judges are reported as rater pairs: agreement on dual-judged cells, concordance of gate trips (exact agreement and Cohen's κ), and a rubric-only baseline (a logistic classifier on criterion text and structural flags with item-grouped five-fold cross-validation over 11,430 labels).

Models, administration and census. Sixteen models sealed before any result; fifteen measured; fourteen on the board (August 2026) over three never-merged routes; one never served; one withdrawn between arms; one re-pinned to the metered route and reported in its own frame. The repeat arm (October 2026) re-administered the 120 board items to eight metered models two or three times at the sealed temperatures. Four further arms administered the 160 remaining items twice, the six CLI-route identities twice on the metered route, every board item paraphrased once (a generator rewrote every prose surface with ids, options, keys, tools, labels and rubric references held byte-identical; eight-gram overlap with the original at most 0.25), and the sixteen pressure-removed twins twice. Census-clean means zero unexpected nulls against scheduled denominators; a null is a re-run; a declared model stays in the denominator; no model-grain statement rests on fewer than twelve cells. Two defects found by audit are reported as findings: sealed temperatures not applied on two CLI routes, and an agentic judging path that omitted the model's in-episode record, repaired and followed by a re-judge of 1,056 records whose grades are reported beside the originals. In all, 9,776 measured administrations and 24,518 scored verdicts, plus 96 administrations and 1,215 verdicts in the ablations.

Statistics. Model effects are η^2 of model with items as units and permutation p over item-label shuffles (4,000 shuffles on the board, 2,000 on the other arms); intervals on means are 95 % item bootstraps (2,000 resamples); pairwise separations are reported as a display (non-overlapping intervals) and tested by Holm-corrected paired Wilcoxon over items; proportions carry Wilson 95 % intervals; the say–do rule is symmetric (say = reasons ≥ 0.5, do = predicate passes) with McNemar exact tests on the discordant cells, Haldane-corrected odds ratios with 95 % intervals on the log scale, and Fisher exact tests between item sets; agreement between judges and between administrations is reported as

mean absolute difference, exact agreement and Cohen's κ; concordances are Spearman at the stated n, with Fisher-z 95 % intervals and leave-one-out ranges where reported, and are displays of agreement; the rubric-only baseline is a logistic classifier with item-grouped five-fold cross-validation and AUC. All tests are exploratory under the withdrawal of the confirmatory track; recomputations reproduce every banked cell score within ±0.0005.

Scale and cost. The board: 120 items per model, 1,680 cells, 1,326 judged cells, 2,242 verdicts. The repeat arm: 2,515 of 2,520 cells, 5,702 verdicts. The four further arms: 5,581 of 5,688 cells, 16,574 verdicts. The re-judge: 1,056 records at no cash cost. The ablations: 96 administrations, 1,215 verdicts, $1.80. Metered spend was about $28 for the repeat arm and $190 for the four further arms; the CLI routes ran at subscription cost (Extended Data Table 19). The instrument's rigor cost far more in authoring, validation and audit than in model calls.

Released items. Twelve board items, one per school and rung, chosen by a stated rule (agentic items of the predicate audit's discriminating class with a typical menu; sustained items whose zero share is closest to the school's median; stated items not edited after the board), are released verbatim with their rubrics, keys, behaviour labels, tool manifests, world state and debrief questions, and are retired from ranking use; their shadow twins stay withheld as the contamination baseline.

Ethics and reporting. Scenarios are mundane-professional; no human participants; the expert gold arm's recruitment and ethics review were prepared and the arm was withdrawn before data collection. Models are listed in sealed census order and no tier, order or leaderboard is asserted anywhere.

Data and code availability

The released items, rubrics and labels, the validator rules, the judge template's clauses and output contract, the analysis code and every tracked output from which the figures and tables are built are in the paper folder of the repository; identifiers, counts and labels for all 280 items are released, and item text, twins, commentary, model responses and judge verdict text are withheld under the release policy of the findings paper. [Repository address and archive DOI to be set.]

Author contributions, competing interests, acknowledgements

[To be set with the author line.]

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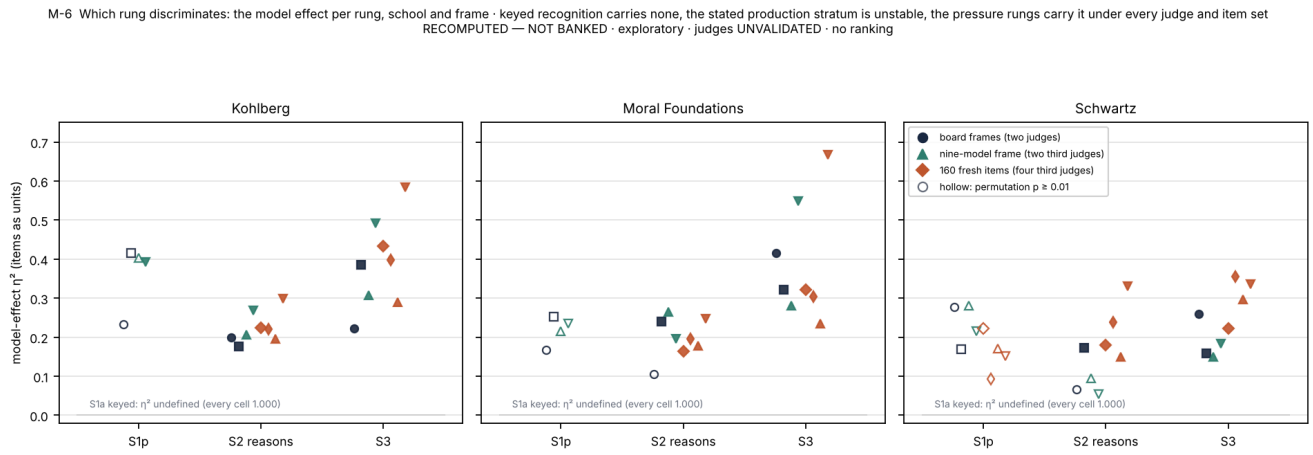
Figure legends

Figure 1. From theory to test. **a**, The three schools as measurement objects: the construct measured on request, its criterion families and the gates written from the theory's own rules against its misreading. **b**, The elicitation ladder: the same construct under increasing cost of fidelity, with what each rung may license and what was observed. **Figure 2.** Recognition saturates, pressure separates, and the gate carries the floor. **a**, Model effect (η^2 of model, items as units) per rung and school on the board's two same-judge frames, the nine-model third-judge frame and the 160 fresh items; hollow markers have permutation $p \geq 0.01$; keyed recognition has no variance. **b**, Band of sustained model-leg means under the sealed reducer and with the gate removed. **c**, Sustained gate trips on the 112 dual-judged cells per school: both judges, one only, neither. **Figure 3.** Saying and doing come apart in a school-specific way. **a**, Probability that an episode with faithful reasons fails the behaviour predicate, per school under four third judges, with the reverse rate, and for the sixteen Schwartz agentic items before and after the competing pressure is removed. **b**, Share of sustained turn criteria not passed at the bare-pressure pole and the new-fact pole, six judges. **c**, Spearman between a model's sustained mean and its agentic behaviour pass rate per school, on the board's items and on fresh items ($n = 7$–8 per point). **Figure 4.** Per-model profiles, the structure between schools, and what capability explains. **a**, **b**, Model-leg means per school and rung in the two board frames (eleven and eight models, sealed census order); agentic behaviour is judge-free and shown where the board's behaviour-versus-reasons table covers the seat. **c**, Pearson correlation of school means across models per frame, with the first principal component's share. **d**, Spearman between a published reasoning scalar and the school's model means per rung, on the board frames, the nine-model frame and the judge-free agentic behaviour; hollow markers $p \geq 0.05$. **Figure 5.** Judges as raters, what repeats, and what the ablations changed. **a**, The evidence-quote compliance screen over eleven candidate judges. **b**, Cohen's κ (exact agreement in brackets) of Kohlberg sustained gate trips among five third judges on identical cells. **c**, Per model, the mean absolute change of a judged sustained cell between administrations under two judges, and the κ of the mechanical agentic channel between administrations. **d**, Label agreement with the sealed template when the template's clauses are removed and when the gates are absent from the prompt, per school, and the Schwartz say-not-do rate under five judges on identical episodes. All judged quantities are from unvalidated language-model judges; `RECOMPUTED — NOT BANKED`; no ranking.

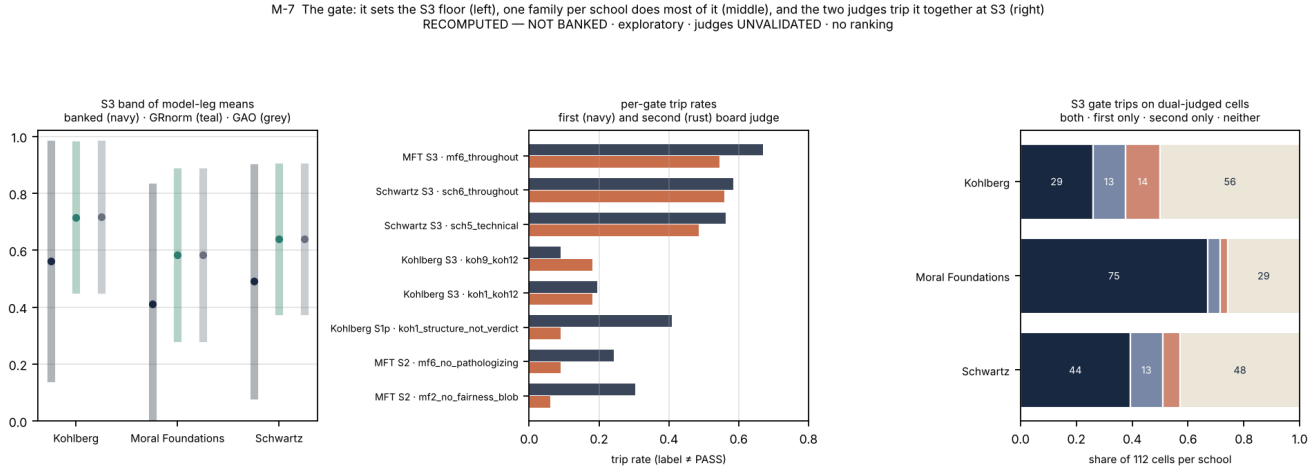
Extended Data

Extended Data Figures

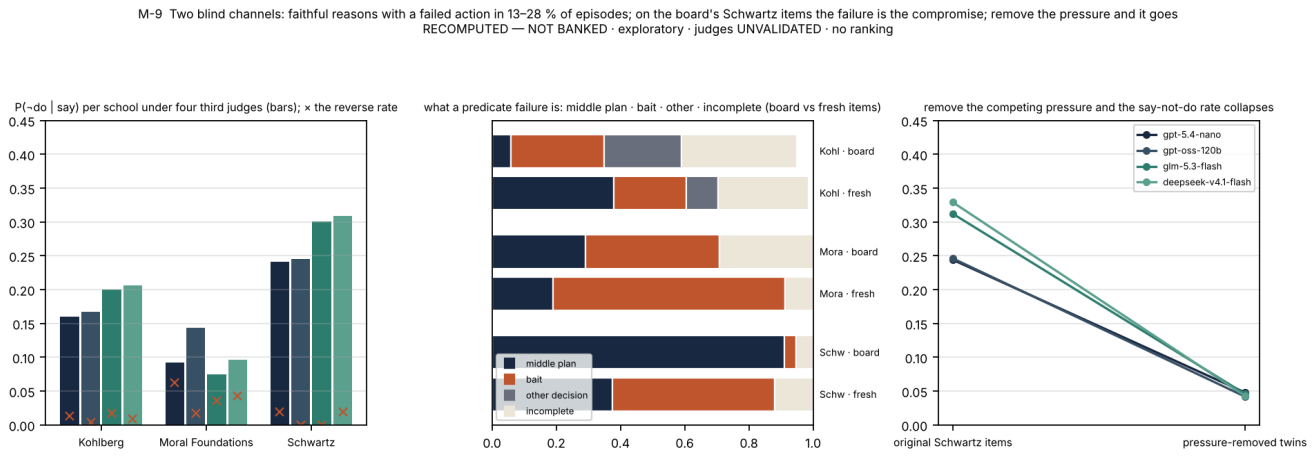
Extended Data Fig. 1. Which rung discriminates: model-effect η^2 per rung, school and frame on the board, the nine-model frame and the 160 fresh items; hollow markers $p \geq 0.01$. **Extended Data Fig. 2.** The gate: sustained bands under the banked reducer, GRnorm and GAO; per-gate trip rates under the two board judges; trips on dual-judged cells. **Extended Data Fig. 3.** Two blind channels: $P(\neg \text{do} \mid \text{say})$ per school under four third judges with the reverse rate; what a predicate failure is made of on board and fresh items; the pressure-removed twins. **Extended Data Fig. 4.** Do the rungs agree about the models: Spearman between a model's sustained mean and its agentic behaviour pass rate per school, by arm and judge. **Extended Data Fig. 5.** Reliability: judged cells between administrations under two judges; the mechanical channel's agreement and κ per model; paraphrase shifts per channel. **Extended Data Fig. 6.** The five metered ablations. All `RECOMPUTED — NOT BANKED`; no ranking.



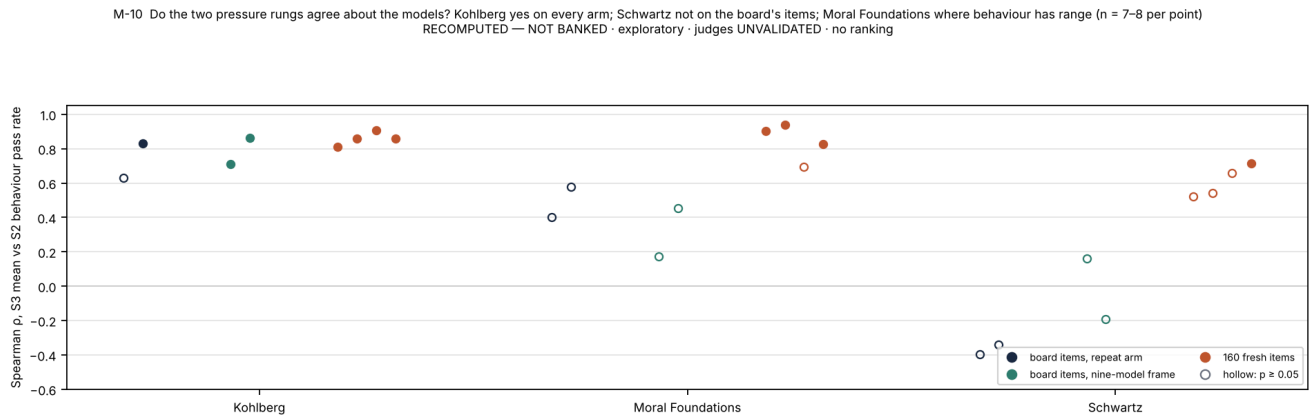
Extended Data Fig. 1 · which rung discriminates Model-effect η² per rung, school and frame on the board, the nine-model frame and the 160 fresh items; hollow markers p ≥ 0.01.



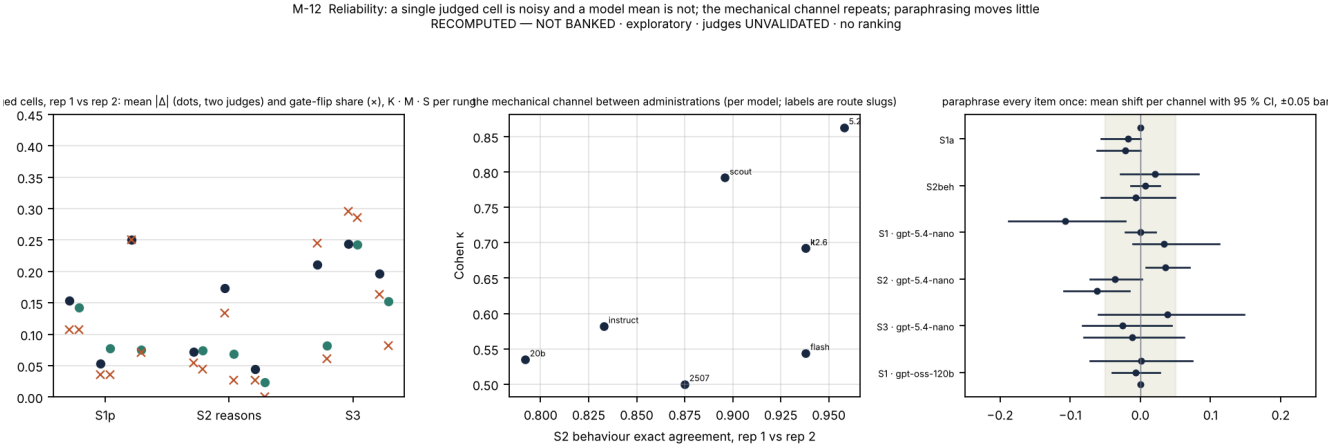
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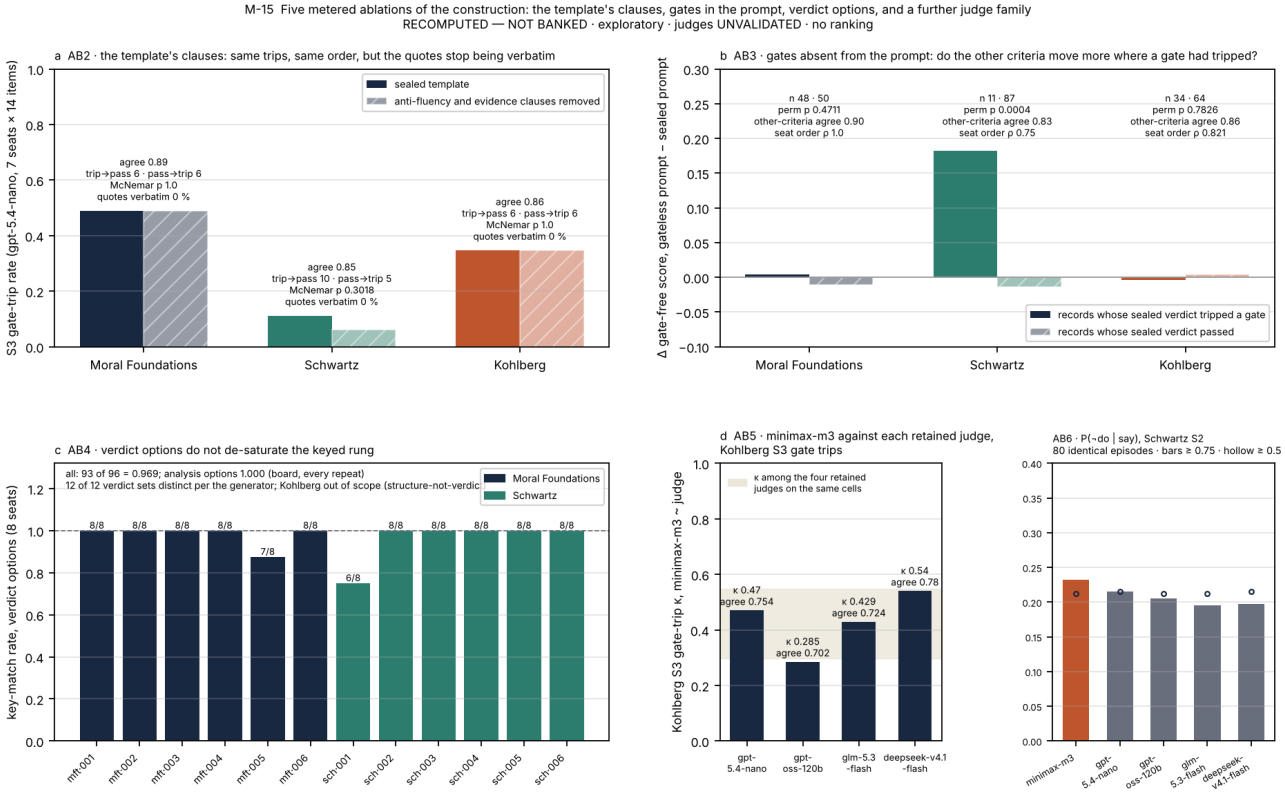
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Extended Data Fig. 4 · do the rungs agree about the models? Spearman between a model's S3 mean and its S2 behaviour pass rate per school, by arm and judge; n = 7–8 per point.



Extended Data Fig. 5 · reliability Judged cells between administrations under two judges; the mechanical channel's agreement and κ per model; paraphrase shifts per channel.



Extended Data Fig. 6 · the five metered ablations AB2 gate-trip rates sealed versus stripped; AB3 anchoring split; AB4 key match per keyed item; AB5/AB6 the fifth judge family.

Extended Data Tables

Extended Data Table 1. Every board model-leg mean by school and rung in the two main board frames (F-A judge gpt-5.6-sol, eleven models; F-B judge claude-opus-5, eight models) and, for the three models judged only in their own frames (F-C, F-D; judge claude-opus-5), in those, with the judge-free agentic behaviour pass rate where the board's behaviour-versus-reasons table covers the seat. Models in sealed census order; frames are never pooled; no ranking.

model (census order)	Kohl S1p F-A · F-B	Kohl S2 reasons F-A · F-B	Kohl S3 F-A · F-B	Kohl S2 behaviour	Mora S1p F-A · F-B	Mora S2 reasons F-A · F-B	Mora S3 F-A · F-B	Mora S2 behaviour	Schw S1p F-A · F-B	Schw S2 reasons F-A · F-B	Schw S3 F-A · F-B	Schw S2 behaviour
gpt-oss-20b	F-D 0.81	F-D 0.87	F-D 0.50	—	F-D 0.50	F-D 0.62	F-D 0.12	—	F-D 0.75	F-D 0.84	F-D 0.41	—
qwen3-235b-thinking	0.78 · 0.61	0.65 · 0.68	0.53 · 0.42	—	0.75 · 0.72	0.84 · 0.62	0.13 · 0.13	—	0.50 · 0.72	0.79 · 0.71	0.09 · 0.09	—
deepseek-v4-flash	0.83 · 0.18	0.89 · 0.93	0.66 · 0.73	—	0.96 · 0.94	0.90 · 0.80	0.32 · 0.23	—	1.00 · 1.00	0.93 · 0.95	0.49 · 0.33	—
mistral-small-3.2-24b	0.70 · 0.22	0.57 · 0.64	0.29 · 0.23	—	0.94 · 0.90	0.79 · 0.78	0.04 · 0.00	—	0.75 · 0.72	0.83 · 0.76	0.47 · 0.35	—
qwen3-8b	0.58 · 0.00	0.70 · 0.69	0.33 · 0.15	—	0.73 · 0.48	0.91 · 0.74	0.00 · 0.00	—	0.75 · 0.50	0.84 · 0.67	0.32 · 0.26	—
gemma-4-31b-it	0.72 · 0.39	0.88 · 0.86	0.66 · 0.49	0.94	0.97 · 0.95	0.90 · 0.85	0.09 · 0.18	1.00	1.00 · 1.00	0.88 · 0.86	0.50 · 0.52	0.69
llama-4-scout	0.72 · 0.58	0.41 · 0.67	0.43 · 0.39	0.38	0.93 · 0.48	0.81 · 0.39	0.37 · 0.21	0.69	1.00 · 0.92	0.90 · 0.87	0.56 · 0.46	0.75
glm-5.2	0.86 · 0.83	0.87 · 0.96	0.83 · 0.97	0.94	0.98 · 0.95	0.98 · 0.99	0.62 · 0.68	1.00	1.00 · 0.75	0.92 · 0.96	0.87 · 0.69	0.56
kimi-k2.6	0.83 · 0.64	0.90 · 0.93	0.70 · 0.81	—	0.75 · 0.73	0.98 · 0.89	0.48 · 0.48	—	1.00 · 0.75	0.94 · 0.94	0.38 · 0.47	—
claude-opus-4-8	0.89 · —	0.98 · —	0.87 · —	1.00	1.00 · —	0.97 · —	0.68 · —	1.00	1.00 · —	0.82 · —	0.39 · —	0.81
claude-sonnet-4-6	0.89 · —	0.92 · —	0.87 · —	0.94	0.98 · —	0.97 · —	0.82 · —	1.00	0.50 · —	0.94 · —	0.89 · —	0.56
claude-haiku-4-5	0.67 · —	0.80 · —	0.81 · —	0.81	0.98 · —	0.96 · —	0.66 · —	0.94	0.75 · —	0.91 · —	0.64 · —	0.69
gpt-5.4	F-C 0.92	F-C 0.43	F-C 0.90	—	F-C 0.95	F-C 0.46	F-C 0.52	—	F-C 1.00	F-C 0.56	F-C 0.58	—
gpt-5.4-mini	F-C 0.86	F-C 0.29	F-C 0.84	—	F-C 0.95	F-C 0.35	F-C 0.40	—	F-C 1.00	F-C 0.47	F-C 0.51	—

Extended Data Table 2. Separation per rung and frame on the board, including keyed recognition: CI-separated pairs, Holm-corrected paired Wilcoxon pairs, η^2 and permutation p.

frame · school	setting	CI-separated pairs / pairs	pairs separated by Holm-corrected paired Wilcoxon (p < 0.05)	model-effect η^2	permutation p	items
F-A Kohlberg	S1a / S1p / S2 / S3	0/55 · 6/55 · 10/55 · 10/55	0 · 0 · 2 · 0	— · 0.233 · 0.199 · 0.222	— · 0.275 · **0.0002** · **0.0002**	6 · 4 · 16 · 14
F-A MFT	S1a / S1p / S2 / S3	0/55 · 0/55 · 9/55 · 25/55	0 · 0 · 0 · 0	— · 0.167 · 0.106 · 0.416	— · 0.889 · 0.027 · **0.0002**	6 · 4 · 16 · 14
F-A Schwartz	S1a / S1p / S2 / S3	0/55 · 0/55 · 0/55 · 18/55	0 · 0 · 0 · 0	— · 0.278 · 0.066 · 0.259	— · 0.450 · 0.309 · **0.0002**	6 · 4 · 16 · 14
F-B Kohlberg	S1a / S1p / S2 / S3	0/28 · 6/28 · 12/28 · 11/28	0 · 0 · 8 · 3	— · 0.415 · 0.177 · 0.385	— · 0.028 · **0.0002** · **0.0002**	6 · 4 · 16 · 14
F-B MFT	S1a / S1p / S2 / S3	0/28 · 0/28 · 10/28 · 12/28	0 · 0 · 3 · 0	— · 0.252 · 0.240 · 0.321	— · 0.249 · **0.0002** · **0.0002**	6 · 4 · 16 · 14
F-B Schwartz	S1a / S1p / S2 / S3	0/28 · 2/28 · 8/28 · 4/28	0 · 0 · 0 · 0	— · 0.170 · 0.173 · 0.159	— · 0.624 · **0.0025** · **0.0017**	6 · 4 · 16 · 14

Extended Data Table 3. The repeat arm per model: sustained mean under two third judges, agentic behaviour pass rate, and the say-not-do rate, per school.

model (census order)	Kohlberg S3 nano / oss	Kohlberg S2 P(do)	Kohlberg P(¬do | say) nano / oss	MFT S3 nano / oss	MFT S2 P(do)	MFT P(¬do | say) nano / oss	Schwartz S3 nano / oss	Schwartz S2 P(do)	Schwartz P(¬do | say) nano / oss
qwen3-235b-thinking	0.73 / 0.87	0.81	0.20 / 0.19	0.30 / 0.36	1.00	0.00 / 0.00	0.64 / 0.81	0.75	0.25 / 0.25
deepseek-v4-flash	0.71 / 0.89	0.94	0.06 / 0.06	0.39 / 0.46	1.00	0.00 / 0.00	0.67 / 0.82	0.85	0.15 / 0.15
mistral-small-3.2-24b	0.21 / 0.40	0.71	0.20 / 0.22	0.13 / 0.17	0.62	0.28 / 0.33	0.65 / 0.72	0.92	0.05 / 0.09
gemma-4-31b-it	0.61 / 0.88	0.92	0.06 / 0.08	0.20 / 0.45	1.00	0.00 / 0.00	0.59 / 0.86	0.73	0.27 / 0.27
llama-4-scout	0.38 / 0.49	0.40	0.50 / 0.50	0.35 / 0.40	0.42	0.46 / 0.59	0.71 / 0.72	0.65	0.29 / 0.29
glm-5.2	0.81 / 0.99	0.94	0.06 / 0.06	0.82 / 0.96	1.00	0.00 / 0.00	0.94 / 0.97	0.50	0.50 / 0.50
kimi-k2.6	0.79 / 0.96	0.91	0.09 / 0.09	0.74 / 0.77	1.00	0.00 / 0.00	0.91 / 0.95	0.75	0.25 / 0.25

Extended Data Table 4. What repeats, per model: keyed recognition, agentic behaviour (exact agreement and κ) and judged sustained cells between administrations, with the August-versus-October κ for the seats administered on both occasions.

seat (census order)	cells rep1 / rep2 / rep3	S1a exact agreement rep1 vs rep2	S2 behaviour exact / κ, rep1 vs rep2	S2 behaviour exact / κ, rep1 vs rep3	κ August vs rep1 (B2 seats)	S3 judged |Δ| · gate flips, rep1 vs rep2 (nano)	same, oss-120b
gpt-oss-20b	120 / 120 / 120	1.000	0.79 / 0.535	0.77 / 0.516	—	0.160 · 7/41 (glm)	—
qwen3-235b-thinking	120 / 120 / 0	1.000	0.88 / 0.5	—	—	0.251 · 11/42	0.181 · 7/42
deepseek-v4-flash	120 / 120 / 120	1.000	0.94 / 0.544	0.92 / 0.458	—	0.242 · 11/42	0.190 · 7/42
mistral-small-3.2-24b	120 / 120 / 120	1.000	0.83 / 0.582	0.77 / 0.413	—	0.250 · 15/42	0.195 · 9/42
gemma-4-31b-it	120 / 120 / 120	1.000	0.94 / 0.692	0.94 / 0.692	0.897	0.171 · 8/42	0.197 · 8/42
llama-4-scout	120 / 120 / 120	1.000	0.90 / 0.792	0.83 / 0.667	0.412	0.211 · 10/42	0.181 · 6/42
glm-5.2	120 / 120 / 0	1.000	0.96 / 0.863	—	0.786	0.165 · 6/42	0.037 · 1/42
kimi-k2.6	120 / 120 / 0	1.000	0.94 / 0.692	—	—	0.230 · 8/42	0.133 · 4/42

Extended Data Table 5. The ladder without the gate: spread and band of model means under the banked reducer, GRnorm and GAO per rung and school, and the Spearman between banked and gate-free model means per frame.

school	setting	banked: SD · min–max	GRnorm: SD · min–max	GAO: SD · min–max	Spearman banked vs GRnorm, F-A · F-B
Kohlberg	S1p	0.246 · 0.00–0.92	0.088 · 0.56–0.89	0.094 · 0.55–0.92	0.62 (n = 11) · 0.69 (n = 8)
Kohlberg	S2	0.189 · 0.29–0.98	0.167 · 0.32–0.97	0.149 · 0.38–0.98	0.81 (n = 11) · 0.98 (n = 8)
Kohlberg	S3	0.238 · 0.15–0.97	0.167 · 0.46–0.97	0.167 · 0.46–0.97	0.94 (n = 11) · 0.98 (n = 8)
MFT	S1p	0.169 · 0.48–1.00	0.048 · 0.78–1.00	0.060 · 0.77–1.00	0.25 (n = 11) · 0.52 (n = 8)
MFT	S2	0.188 · 0.35–0.99	0.124 · 0.54–0.98	0.114 · 0.57–0.99	0.90 (n = 11) · 0.88 (n = 8)
MFT	S3	0.253 · 0.00–0.82	0.186 · 0.29–0.88	0.186 · 0.29–0.88	0.91 (n = 11) · 0.96 (n = 8)
Schwartz	S1p	0.176 · 0.50–1.00	0.024 · 0.90–1.00	0.035 · 0.87–1.00	0.51 (n = 11) · -0.10 (n = 8)
Schwartz	S2	0.127 · 0.47–0.96	0.102 · 0.54–0.95	0.090 · 0.59–0.96	0.80 (n = 11) · 0.83 (n = 8)
Schwartz	S3	0.196 · 0.09–0.89	0.142 · 0.39–0.89	0.142 · 0.39–0.89	0.71 (n = 11) · 0.82 (n = 8)

Extended Data Table 6. The say–do rates under three threshold rules (strict, symmetric 0.5, symmetric 0.75) per school and judge.

school	judge	strict (say ≥ 0.75, not-say < 0.5): P(¬do | say) / P(¬say | do) / p (episodes in the excluded band)	symmetric 0.5: same	symmetric 0.75: same
Kohlberg	gpt-5.4-nano	0.16 / 0.01 / < 10^{-4} (band 16)	0.19 / 0.01 / < 10^{-4}	0.16 / 0.03 / < 10^{-4}
Kohlberg	gpt-oss-120b	0.17 / 0.00 / < 10^{-4} (band 13)	0.20 / 0.00 / < 10^{-4}	0.17 / 0.01 / < 10^{-4}
Kohlberg	glm-5.3-flash	0.20 / 0.02 / < 10^{-4} (band 13)	0.26 / 0.02 / < 10^{-4}	0.20 / 0.03 / < 10^{-4}
Kohlberg	deepseek-v4.1-flash	0.21 / 0.01 / < 10^{-4} (band 11)	0.24 / 0.01 / < 10^{-4}	0.21 / 0.03 / < 10^{-4}
MFT	gpt-5.4-nano	0.09 / 0.07 / 0.324 (band 25)	0.14 / 0.06 / 0.0032	0.09 / 0.10 / 0.7709

school	judge	strict (say ≥ 0.75, not-say < 0.5): P(¬do | say) / P(¬say | do) / p (episodes in the excluded band)	symmetric 0.5: same	symmetric 0.75: same
MFT	gpt-oss-120b	0.14 / 0.02 / < 10^{-4} (band 3)	0.15 / 0.02 / < 10^{-4}	0.14 / 0.02 / < 10^{-4}
MFT	glm-5.3-flash	0.07 / 0.04 / 0.2101 (band 5)	0.11 / 0.04 / 0.0266	0.07 / 0.04 / 0.2101
MFT	deepseek-v4.1-flash	0.10 / 0.04 / 0.1153 (band 5)	0.11 / 0.04 / 0.0347	0.10 / 0.06 / 0.2863
Schwartz	gpt-5.4-nano	0.24 / 0.02 / < 10^{-4} (band 1)	0.24 / 0.02 / < 10^{-4}	0.24 / 0.02 / < 10^{-4}
Schwartz	gpt-oss-120b	0.24 / 0.00 / < 10^{-4} (band 3)	0.25 / 0.00 / < 10^{-4}	0.24 / 0.01 / < 10^{-4}
Schwartz	glm-5.3-flash	0.30 / 0.00 / < 10^{-4} (band 4)	0.31 / 0.00 / < 10^{-4}	0.30 / 0.01 / < 10^{-4}
Schwartz	deepseek-v4.1-flash	0.31 / 0.02 / < 10^{-4} (band 6)	0.33 / 0.02 / < 10^{-4}	0.31 / 0.03 / < 10^{-4}

Extended Data Table 7. The pressure-removed twins of the sixteen Schwartz agentic items: behaviour, P(say), say-not-do with Wilson intervals, Fisher exact tests, and the failure classes before and after.

judge (reasons)	episodes original / twin	P(do) original / twin	P(say) original / twin	original: say-not-do n (P(¬do | say) [Wilson 95%])	twin: say-not-do n (P(¬do | say) [Wilson 95%])	do-not-say original / twin	Fisher exact p, say-not-do share (original vs twin)	say-not-do classes (strict rule) middle / bait: original → twin
gpt-5.4-nano	288 / 224	0.75 / 0.94	0.97 / 0.92	68 (0.24 [0.20, 0.30])	10 (0.05 [0.03, 0.09])	4 / 13	$< 10^{-4}$	65 / 2 → 9 / 0
gpt-oss-120b	288 / 224	0.75 / 0.94	0.99 / 0.95	70 (0.25 [0.20, 0.30])	9 (0.04 [0.02, 0.08])	0 / 5	$< 10^{-4}$	66 / 2 → 9 / 0
glm-5.3-flash	160 / 256	0.69 / 0.94	1.00 / 0.93	50 (0.31 [0.25, 0.39])	11 (0.05 [0.03, 0.08])	0 / 13	$< 10^{-4}$	45 / 0 → 11 / 0
deepseek-v4.1-flash	160 / 256	0.66 / 0.94	0.97 / 0.90	51 (0.33 [0.26, 0.41])	10 (0.04 [0.02, 0.08])	2 / 20	$< 10^{-4}$	44 / 0 → 7 / 0
(behaviour, judge-free)	352 / 288	0.71 [0.66, 0.76] / 0.95 [0.92, 0.97]	—	—	—	—	—	classes original {'middle': 92, 'faithful': 250, 'other_or_incomplete': 7, 'bait': 3} → twin {'faithful': 273, 'other_or_incomplete': 4, 'middle': 11}

Extended Data Table 8. The 160 fresh items: model-effect panels per school, rung and judge, with the complete-case variant.

school	setting	judge	items common to all models / models	CI-separated pairs / pairs	η² (model)	permutation p	model means: min – max (median)	complete-case variant without llama-4-scout: items / models: η², p
Kohlberg	S2	gpt-5.4-nano	18 / 7	5 / 21	0.225	0.0005	0.67 – 0.95 (median 0.95)	18 / 6: η² 0.078, p 0.049
Kohlberg	S2	gpt-oss-120b	18 / 7	9 / 21	0.221	0.0005	0.80 – 1.00 (median 0.99)	18 / 6: η² 0.220, p 0.0005
Kohlberg	S2	glm-5.3-flash	18 / 8	5 / 28	0.196	0.001	0.73 – 0.99 (median 0.97)	18 / 7: η² 0.053, p 0.2334
Kohlberg	S2	deepseek-v4.1-flash	14 / 8	11 / 28	0.299	0.0005	0.65 – 0.95 (median 0.90)	16 / 7: η² 0.274, p 0.0005
Kohlberg	S3	gpt-5.4-nano	18 / 7	12 / 21	0.434	0.0005	0.07 – 0.83 (median 0.53)	18 / 6: η² 0.368, p 0.0005
Kohlberg	S3	gpt-oss-120b	18 / 7	10 / 21	0.397	0.0005	0.29 – 0.91 (median 0.72)	18 / 6: η² 0.334, p 0.0005
Kohlberg	S3	glm-5.3-flash	18 / 8	11 / 28	0.289	0.0005	0.21 – 0.89 (median 0.58)	18 / 7: η² 0.213, p 0.0005
Kohlberg	S3	deepseek-v4.1-flash	18 / 8	14 / 28	0.584	0.0005	0.03 – 0.79 (median 0.20)	18 / 7: η² 0.560, p 0.0005
MFT	S2	gpt-5.4-nano	21 / 7	6 / 21	0.164	0.0005	0.42 – 0.86 (median 0.69)	21 / 6: η² 0.107, p 0.001
MFT	S2	gpt-oss-120b	21 / 7	7 / 21	0.196	0.0005	0.63 – 0.99 (median 0.93)	21 / 6: η² 0.144, p 0.0005
MFT	S2	glm-5.3-flash	21 / 8	10 / 28	0.178	0.001	0.63 – 1.00 (median 0.86)	21 / 7: η² 0.165, p 0.0005
MFT	S2	deepseek-v4.1-flash	21 / 8	11 / 28	0.248	0.0005	0.49 – 0.97 (median 0.75)	21 / 7: η² 0.220, p 0.0005
MFT	S3	gpt-5.4-nano	20 / 7	9 / 21	0.322	0.0005	0.12 – 0.82 (median 0.30)	21 / 6: η² 0.329, p 0.0005
MFT	S3	gpt-oss-120b	20 / 7	9 / 21	0.304	0.0005	0.29 – 0.96 (median 0.50)	21 / 6: η² 0.286, p 0.0005
MFT	S3	glm-5.3-flash	18 / 8	13 / 28	0.235	0.0005	0.37 – 0.99 (median 0.68)	19 / 7: η² 0.254, p 0.0005
MFT	S3	deepseek-v4.1-flash	18 / 8	17 / 28	0.667	0.0005	0.00 – 0.94 (median 0.07)	19 / 7: η² 0.648, p 0.0005
Schwartz	S2	gpt-5.4-nano	36 / 7	10 / 21	0.180	0.0005	0.37 – 0.89 (median 0.73)	36 / 6: η² 0.121, p 0.0005
Schwartz	S2	gpt-oss-120b	36 / 7	9 / 21	0.238	0.0005	0.64 – 0.98 (median 0.94)	36 / 6: η² 0.141, p 0.0005
Schwartz	S2	glm-5.3-flash	36 / 8	10 / 28	0.150	0.0005	0.54 – 0.96 (median 0.76)	36 / 7: η² 0.144, p 0.0005
Schwartz	S2	deepseek-v4.1-flash	17 / 8	15 / 28	0.331	0.0005	0.37 – 0.92 (median 0.60)	18 / 7: η² 0.258, p 0.0005
Schwartz	S3	gpt-5.4-nano	12 / 7	3 / 21	0.222	0.0025	0.18 – 0.72 (median 0.41)	36 / 6: η² 0.121, p 0.0005
Schwartz	S3	gpt-oss-120b	12 / 7	8 / 21	0.356	0.0005	0.35 – 0.95 (median 0.76)	36 / 6: η² 0.149, p 0.0005
Schwartz	S3	glm-5.3-flash	12 / 8	9 / 28	0.296	0.0005	0.30 – 0.96 (median 0.56)	35 / 7: η² 0.164, p 0.0005

school	setting	judge	items common to all models / models	CI-separated pairs / pairs	η^2 (model)	permutation p	model means: min – max (median)	complete-case variant without llama-4-scout: items / models: η^2, p
Schwartz	S3	deepseek-v4.1-flash	12 / 8	7 / 28	0.336	0.0005	0.09 – 0.84 (median 0.29)	34 / 7: η^2 0.212, p 0.0005

Extended Data Table 9. The nine-model frame under two further third judges: η^2, separated pairs and seat-mean ranges per school and rung.

judge	school	S1p η^2 / p (SMALL)	S2 η^2 / p · separated pairs of 28	S3 η^2 / p · separated pairs	S3 seat-mean min / median / max
glm-5.3-flash	Kohlberg	0.403 / 0.025	0.207 / 0.001 · 13	0.308 / 0.0005 · 10	0.317 / 0.777 / 0.978
glm-5.3-flash	MFT	0.215 / 0.5727	0.265 / 0.0005 · 15	0.281 / 0.0005 · 9	0.196 / 0.483 / 0.979
glm-5.3-flash	Schwartz	0.28 / 0.5177	0.095 / 0.093 · 3	0.15 / 0.0005 · 8	0.675 / 0.829 / 0.986
deepseek-v4.1-flash	Kohlberg	0.392 / 0.0005	0.268 / 0.0005 · 12	0.492 / 0.0005 · 18	0.188 / 0.696 / 0.995
deepseek-v4.1-flash	MFT	0.235 / 0.5932	0.196 / 0.0015 · 10	0.548 / 0.0005 · 17	0.0 / 0.178 / 0.859
deepseek-v4.1-flash	Schwartz	0.216 / 0.2624	0.054 / 0.3438 · 4	0.183 / 0.0005 · 5	0.455 / 0.714 / 0.895

Extended Data Table 10. The two poles: share of sustained turn criteria not passed by class (bare pressure, authority or social pressure, new fact, gates, other) per school, arm and judge.

arm	judge	school	bare pressure: not passed (labels)	authority / social pressure	new fact	gates	other turn criteria
board	claude-opus-5	Kohlberg	0.33 (209)	0.48 (33)	0.34 (165)	0.31 (154)	0.50 (275)
board	claude-opus-5	Moral Foundations	0.74 (232)	0.50 (121)	0.32 (178)	0.67 (154)	0.41 (165)
board	claude-opus-5	Schwartz	0.55 (153)	0.70 (88)	0.30 (153)	0.46 (153)	0.46 (273)
board	gpt-5.6-sol	Kohlberg	0.21 (209)	0.33 (33)	0.29 (165)	0.30 (154)	0.41 (275)
board	gpt-5.6-sol	Moral Foundations	0.67 (231)	0.59 (121)	0.23 (176)	0.55 (154)	0.32 (165)
board	gpt-5.6-sol	Schwartz	0.51 (154)	0.58 (88)	0.19 (154)	0.41 (154)	0.35 (275)
repeat arm / nine-model frame	deepseek-v4.1-flash	Kohlberg	0.21 (184)	0.44 (27)	0.26 (147)	0.28 (137)	0.47 (247)
repeat arm / nine-model frame	deepseek-v4.1-flash	Moral Foundations	0.68 (209)	0.58 (109)	0.18 (159)	0.58 (139)	0.24 (149)
repeat arm / nine-model frame	deepseek-v4.1-flash	Schwartz	0.36 (139)	0.70 (80)	0.14 (139)	0.23 (139)	0.30 (248)
repeat arm / nine-model frame	glm-5.3-flash	Kohlberg	0.17 (184)	0.41 (27)	0.15 (147)	0.20 (137)	0.42 (247)
repeat arm / nine-model frame	glm-5.3-flash	Moral Foundations	0.55 (209)	0.35 (109)	0.09 (159)	0.42 (139)	0.04 (149)
repeat arm / nine-model frame	glm-5.3-flash	Schwartz	0.25 (139)	0.46 (80)	0.07 (139)	0.09 (139)	0.20 (248)
repeat arm / nine-model frame	gpt-5.4-nano	Kohlberg	0.22 (342)	0.41 (54)	0.17 (270)	0.35 (252)	0.35 (450)
repeat arm / nine-model frame	gpt-5.4-nano	Moral Foundations	0.64 (378)	0.32 (198)	0.19 (288)	0.52 (252)	0.13 (270)
repeat arm / nine-model frame	gpt-5.4-nano	Schwartz	0.26 (252)	0.56 (144)	0.08 (252)	0.17 (252)	0.27 (450)
repeat arm / nine-model frame	gpt-oss-120b	Kohlberg	0.14 (342)	0.20 (54)	0.08 (270)	0.16 (252)	0.23 (450)
repeat arm / nine-model frame	gpt-oss-120b	Moral Foundations	0.56 (378)	0.23 (198)	0.12 (288)	0.41 (252)	0.07 (270)

arm	judge	school	bare pressure: not passed (labels)	authority / social pressure	new fact	gates	other turn criteria
repeat arm / nine-model frame	gpt-oss-120b	Schwartz	0.15 (252)	0.26 (144)	0.05 (252)	0.06 (252)	0.22 (450)

Extended Data Table 11. Ablation AB1: the sustained rung re-reduced under hold-firm-only and new-fact-only rubrics, per frame and school.

frame	school	banked rule: SD of model means · η²	hold-firm-only rubric: SD · η² · Spearman with banked	new-fact-only rubric: SD · η² · Spearman with banked	models
F-A	Kohlberg	0.203 · η² 0.23	0.219 · η² 0.26 · ρ 1.00	0.199 · η² 0.22 · ρ 0.99	11
F-A	Moral Foundations	0.277 · η² 0.42	0.271 · η² 0.42 · ρ 0.99	0.286 · η² 0.38 · ρ 0.98	11
F-A	Schwartz	0.222 · η² 0.26	0.217 · η² 0.26 · ρ 0.99	0.226 · η² 0.24 · ρ 0.97	11
F-B	Kohlberg	0.274 · η² 0.40	0.287 · η² 0.44 · ρ 1.00	0.279 · η² 0.42 · ρ 1.00	8
F-B	Moral Foundations	0.219 · η² 0.32	0.227 · η² 0.34 · ρ 0.98	0.215 · η² 0.28 · ρ 0.93	8
F-B	Schwartz	0.169 · η² 0.16	0.169 · η² 0.17 · ρ 0.98	0.175 · η² 0.16 · ρ 0.98	8

Extended Data Table 12. Paraphrase sensitivity: mean shift per channel, judge and school with 95 % intervals.

channel · judge	school	items	mean shift paraphrase − original [95% CI]	CI within ±0.05	strong paraphrases: items (mean shift)	items flagged
S1a	Kohlberg	6	+0.000 [+0.000, +0.000]	yes	6 (+0.000)	0
S1a	MFT	6	-0.018 [-0.056, +0.000]	no	4 (+0.000)	0
S1a	Schwartz	6	-0.021 [-0.062, +0.000]	no	6 (-0.021)	0
S2beh	Kohlberg	16	+0.021 [-0.028, +0.083]	no	8 (+0.014)	1
S2beh	MFT	16	+0.007 [-0.014, +0.028]	yes	4 (+0.028)	0
S2beh	Schwartz	16	-0.007 [-0.056, +0.049]	no	5 (+0.000)	0
S1 · gpt-5.4-nano	Kohlberg	4	-0.107 [-0.188, -0.021]	no	4 (-0.107)	0
S1 · gpt-5.4-nano	MFT	4	+0.000 [-0.021, +0.022]	yes	4 (+0.000)	0
S1 · gpt-5.4-nano	Schwartz	4	+0.034 [-0.011, +0.112]	no	4 (+0.034)	0
S2 · gpt-5.4-nano	Kohlberg	16	+0.036 [+0.008, +0.070]	no	8 (+0.016)	0
S2 · gpt-5.4-nano	MFT	16	-0.036 [-0.072, +0.002]	no	4 (-0.025)	0
S2 · gpt-5.4-nano	Schwartz	16	-0.062 [-0.109, -0.016]	no	5 (-0.120)	1
S3 · gpt-5.4-nano	Kohlberg	14	+0.039 [-0.060, +0.148]	no	14 (+0.039)	2
S3 · gpt-5.4-nano	MFT	14	-0.025 [-0.083, +0.044]	no	9 (-0.029)	1
S3 · gpt-5.4-nano	Schwartz	14	-0.012 [-0.081, +0.062]	no	8 (-0.061)	1
S1 · gpt-oss-120b	Kohlberg	4	+0.001 [-0.072, +0.074]	no	4 (+0.001)	0
S1 · gpt-oss-120b	MFT	4	-0.007 [-0.040, +0.028]	yes	4 (-0.007)	0
S1 · gpt-oss-120b	Schwartz	4	+0.000 [+0.000, +0.000]	yes	4 (+0.000)	0
S2 · gpt-oss-120b	Kohlberg	16	+0.025 [+0.000, +0.055]	no	8 (+0.007)	0

channel · judge	school	items	mean shift paraphrase − original [95% CI]	CI within ±0.05	strong paraphrases: items (mean shift)	items flagged
S2 · gpt-oss-120b	MFT	16	-0.017 [-0.060, +0.027]	no	4 (-0.032)	0
S2 · gpt-oss-120b	Schwartz	16	-0.031 [-0.063, +0.000]	no	5 (-0.041)	0
S3 · gpt-oss-120b	Kohlberg	14	-0.010 [-0.048, +0.021]	yes	14 (-0.010)	0
S3 · gpt-oss-120b	MFT	14	-0.032 [-0.111, +0.060]	no	9 (-0.017)	2
S3 · gpt-oss-120b	Schwartz	14	-0.025 [-0.094, +0.047]	no	8 (-0.099)	2
S1 · glm-5.3-flash	Kohlberg	4	+0.119 [+0.027, +0.210]	no	4 (+0.119)	0
S1 · glm-5.3-flash	MFT	4	+0.008 [-0.019, +0.043]	yes	4 (+0.008)	0
S1 · glm-5.3-flash	Schwartz	4	+0.104 [+0.000, +0.207]	no	4 (+0.104)	1
S2 · glm-5.3-flash	Kohlberg	16	+0.075 [+0.043, +0.109]	no	8 (+0.063)	0
S2 · glm-5.3-flash	MFT	16	+0.001 [-0.038, +0.044]	yes	4 (+0.014)	0
S2 · glm-5.3-flash	Schwartz	16	-0.022 [-0.050, +0.003]	yes	5 (-0.015)	0
S3 · glm-5.3-flash	Kohlberg	14	+0.082 [+0.021, +0.160]	no	14 (+0.082)	1
S3 · glm-5.3-flash	MFT	14	+0.023 [-0.062, +0.099]	no	9 (+0.028)	1
S3 · glm-5.3-flash	Schwartz	14	-0.034 [-0.083, +0.015]	no	8 (-0.061)	0
S1 · deepseek-v4.1-flash	Kohlberg	4	-0.062 [-0.103, -0.022]	no	4 (-0.062)	0
S1 · deepseek-v4.1-flash	MFT	4	+0.006 [-0.006, +0.019]	yes	4 (+0.006)	0
S1 · deepseek-v4.1-flash	Schwartz	4	-0.020 [-0.138, +0.097]	no	4 (-0.020)	0
S2 · deepseek-v4.1-flash	Kohlberg	16	+0.033 [-0.001, +0.069]	no	8 (+0.016)	0
S2 · deepseek-v4.1-flash	MFT	16	-0.014 [-0.056, +0.026]	no	4 (+0.020)	0
S2 · deepseek-v4.1-flash	Schwartz	16	-0.017 [-0.053, +0.020]	no	5 (-0.016)	0
S3 · deepseek-v4.1-flash	Kohlberg	14	-0.005 [-0.073, +0.054]	no	14 (-0.005)	1
S3 · deepseek-v4.1-flash	MFT	14	+0.000 [-0.049, +0.047]	yes	9 (+0.005)	0
S3 · deepseek-v4.1-flash	Schwartz	14	-0.043 [-0.133, +0.030]	no	8 (-0.126)	3

Extended Data Table 13. Length control: model-level, within-item and within-model correlations between response length and judged score per judge and rung, and for the mechanical channel.

judge	setting	cells	seats	model-level Spearman ρ (mean length vs mean score)	within-item median ρ	within-seat median ρ
gpt-5.4-nano	S3	294	7	0.14 (p 0.76)	0.2 (62% of 42 items positive)	-0.04
gpt-5.4-nano	S2	336	7	0.45 (p 0.31)	0.32 (73% of 37 items positive)	0.02
gpt-oss-120b	S3	294	7	0.5 (p 0.253)	0.24 (69% of 39 items positive)	0.03

judge	setting	cells	seats	model-level Spearman ρ (mean length vs mean score)	within-item median ρ	within-seat median ρ
gpt-oss-120b	S2	336	7	0.61 (p 0.148)	0.41 (85% of 26 items positive)	0.03
glm-5.3-flash	S3	335	8	0.24 (p 0.57)	0.11 (62% of 42 items positive)	-0.03
glm-5.3-flash	S2	384	8	0.0 (p 1.0)	0.1 (74% of 39 items positive)	0.21
deepseek-v4.1-flash-1017	S3	335	8	0.4 (p 0.32)	0.26 (74% of 42 items positive)	0.03
deepseek-v4.1-flash-1017	S2	384	8	-0.02 (p 0.955)	0.3 (75% of 44 items positive)	0.09
— (mechanical)	S2 behaviour		9	-0.25 (p 0.509)	—	—

Extended Data Table 14. Capability concordance on the nine-model frame: Spearman with a published reasoning scalar and an agentic scalar, with Fisher-z intervals and leave-one-out ranges.

school	S3 under glm vs MMLU-Pro-class	S3 under deepseek vs MMLU-Pro-class	S2 behaviour (mechanical) vs MMLU-Pro-class	say-do rate (nano) vs MMLU-Pro-class	say-do rate (glm) vs MMLU-Pro-class	S3 under glm vs SWE-bench-class	S2 behaviour vs SWE-bench-class
Kohlberg	+0.93 [+0.65, +0.99] LOO +0.89…+1.00 (n 8)	+0.90 [+0.55, +0.98] LOO +0.86…+0.96 (n 8)	+0.84 [+0.40, +0.97] LOO +0.76…+0.85 (n 9)	-0.71 [-0.95, +0.09] LOO -0.88…-0.58 (n 7)	-0.88 [-0.98, -0.46] LOO -0.93…-0.82 (n 8)	+0.82 [+0.18, +0.97] LOO +0.71…+0.89 (n 7)	+0.87 [+0.44, +0.98] LOO +0.80…+0.91 (n 8)
MFT	+0.79 [+0.18, +0.96] LOO +0.68…+0.89 (n 8)	+0.79 [+0.18, +0.96] LOO +0.68…+0.89 (n 8)	+0.59 [-0.12, +0.90] LOO +0.41…+0.73 (n 9)	-0.76 [-0.96, -0.01] LOO -0.78…-0.65 (n 7)	-0.79 [-0.96, -0.20] LOO -0.80…-0.76 (n 8)	+0.61 [-0.27, +0.93] LOO +0.37…+0.94 (n 7)	+0.58 [-0.21, +0.91] LOO +0.61…+0.61 (n 8)
Schwartz	+0.93 [+0.65, +0.99] LOO +0.89…+1.00 (n 8)	+0.86 [+0.39, +0.97] LOO +0.79…+0.89 (n 8)	+0.03 [-0.65, +0.68] LOO -0.21…+0.48 (n 9)	+0.29 [-0.59, +0.86] LOO -0.14…+0.49 (n 7)	-0.10 [-0.75, +0.65] LOO -0.67…+0.11 (n 8)	+0.93 [+0.58, +0.99] LOO +0.89…+0.94 (n 7)	+0.22 [-0.57, +0.80] LOO -0.11…+0.64 (n 8)

Extended Data Table 15. Ablation AB2: the sealed judge template against the template without its anti-fluency and evidence clauses, same judge, same 294 cells.

school	cells	label agreement with the sealed template: all · gates · other criteria	gate-trip rate, sealed → stripped (trip→pass / pass→trip; McNemar p)	banked score mean, sealed → stripped (paired Δ)	gate-free mean, sealed → stripped (paired Δ)	seat order kept (Spearman, 7 seats)	score–length Spearman, sealed → stripped	owed quotes still given · verbatim
Moral Foundations	98	0.89 · 0.87 · 0.89	0.49 → 0.49 (6 / 6; p 1.0)	0.42 → 0.43 (+0.003)	0.70 → 0.69 (-0.014)	0.964	0.14 → 0.17	94 % · 0 %
Schwartz	98	0.85 · 0.85 · 0.84	0.11 → 0.06 (10 / 5; p 0.3018)	0.77 → 0.80 (+0.029)	0.83 → 0.84 (+0.003)	0.821	-0.19 → -0.15	90 % · 0 %
Kohlberg	98	0.86 · 0.88 · 0.86	0.35 → 0.35 (6 / 6; p 1.0)	0.59 → 0.60 (+0.005)	0.79 → 0.78 (-0.006)	0.964	-0.06 → -0.12	94 % · 0 %

Extended Data Table 16. Ablation AB3: the gateless prompt against the sealed prompt on the other criteria, with the anchoring split.

school	cells	other-criteria label agreement with the sealed template	other-criteria PASS share, sealed → gateless	gate-free mean, sealed → gateless (paired Δ)	seat order kept (Spearman)	records whose sealed verdict tripped a gate: n · mean Δ gate-free	records whose sealed verdict passed: n · mean Δ	permutation p for the difference	other-criteria agreement, tripped · passed
Moral Foundations	98	0.90	0.68 → 0.68	0.70 → 0.70 (-0.004)	1.000	48 · +0.004	50 · -0.011	0.4711	0.88 · 0.92
Schwartz	98	0.83	0.76 → 0.78	0.83 → 0.84 (+0.008)	0.750	11 · +0.182	87 · -0.014	0.0004	0.72 · 0.85
Kohlberg	98	0.86	0.75 → 0.75	0.79 → 0.79 (+0.001)	0.821	34 · -0.004	64 · +0.004	0.7826	0.79 · 0.90

Extended Data Table 17. Ablation AB5: the fifth judge family against each retained third judge on identical Kohlberg sustained cells, with the retained judges' own pairs for scale.

judge pair	cells	score Pearson	mean |Δ|	exactly equal	gate-trip agreement	Cohen's κ	trips only first / only second	seat order, Spearman
minimax-m3 ~ gpt-5.4-nano	252	0.65	0.210	0.53	0.75	0.47	34 / 28	1.00
minimax-m3 ~ gpt-oss-120b	252	0.56	0.276	0.43	0.70	0.28	65 / 10	0.89
minimax-m3 ~ glm-5.3-flash	123	0.60	0.279	0.41	0.72	0.43	32 / 2	1.00
minimax-m3 ~ deepseek-v4.1-flash	123	0.69	0.201	0.50	0.78	0.54	21 / 6	0.96
gpt-5.4-nano~gpt-oss-120b (for scale)	252	0.65	0.213	0.54	0.79	0.47	51 / 2	—
gpt-5.4-nano~glm-5.3-flash (for scale)	84	0.62	0.225	0.50	0.79	0.51	16 / 2	—
gpt-5.4-nano~deepseek-v4.1-flash (for scale)	84	0.64	0.195	0.59	0.77	0.49	12 / 7	—
gpt-oss-120b~glm-5.3-flash (for scale)	84	0.56	0.201	0.46	0.80	0.34	5 / 12	—
gpt-oss-120b~deepseek-v4.1-flash (for scale)	84	0.58	0.222	0.50	0.75	0.30	4 / 17	—
glm-5.3-flash~deepseek-v4.1-flash (for scale)	109	0.69	0.176	0.59	0.81	0.55	4 / 17	—

Extended Data Table 18. Ablation AB6: the Schwartz agentic 2×2 on identical episodes under each reasons judge.

reasons judge (Schwartz S2, identical episodes)	n	reasons mean	P(say), reasons ≥ 0.75	P(say), reasons ≥ 0.5	say-not-do: n · P(¬do | say) at 0.75	at 0.5	do-not-say: n · P(¬say | do)
minimax-m3	80	0.756	0.70	0.82	13 · 0.23	14 · 0.21	11 · 0.17
gpt-5.4-nano	80	0.973	0.99	0.99	17 · 0.21	17 · 0.21	1 · 0.02
gpt-oss-120b	80	0.985	0.97	1.00	16 · 0.20	17 · 0.21	0 · 0.00
glm-5.3-flash	80	0.964	0.96	1.00	15 · 0.20	17 · 0.21	0 · 0.00
deepseek-v4.1-flash	80	0.945	0.95	0.99	15 · 0.20	17 · 0.21	1 · 0.02

Extended Data Table 19. Scale of the work: instrument, roster, board, repeat arm, gap-closure arms, re-judge, audits and ablations, with metered cost.

what	size	cells	verdicts · judges	metered cost
instrument	280 public items · 280 shadow twins · 262 rubrics	S1 38 · S2 123 · S3 119	15 validator gates · 26 amendments · sealed 2026-08-31 (A-25)	—
roster	16 sealed · 15 measured · 14 on the board	3 routes	one never served, one withdrawn between arms	—
board (Aug 2026)	120 items × 14 models	1,680 cells · 1,326 judged	2,242 verdicts · 2 judges · 6 legs · 4 frames	subscription routes + metered
repeat arm (Oct 2026)	8 models × 2 (3 on five models)	2,515 of 2,520 cells	5,702 verdicts · 4 third judges	$27.7
gap-closure arms (Oct 2026)	160 fresh items · six route re-pins · paraphrases · pressure-removed twins	5,581 of 5,688 cells	16,574 verdicts · 4 third judges	$160.01 (+ ≈ $28 judges, $1.58 paraphrases)
re-judge (Oct 2026)	1,056 board S2 records with the written record visible	528 + 1,107 cells	one judge identity per leg	$0
audits	48 predicates over 1,344 episodes · temperature forensics on 5 models	11 judge candidates screened, 5 retained	Gate M panel 600 cells (expert gold withdrawn, A-26)	$0
ablations (1 Nov 2026, §S7)	AB2/AB3 two judge-template variants on 294 S3 cells · AB4 twelve verdict-option items on 8 seats · AB5/AB6 a further judge family on Kohlberg S3 and Schwartz S2	96 administrations · 4 judge candidates smoked	1,215 verdicts · gpt-5.4-nano, minimax-m3	$1.80 (not in the totals below)
totals	—	9,776 measured administrations	24,518 scored judge verdicts	≈ $188 metered